Hunters of the Ice Age: The Biology of Upper Paleolithic People

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ABSTRACT The Upper Paleolithic represents both the phase during which anatomically modern humans appeared and the climax of hunter–gatherer cultures. Demographic expansion into new areas that took place during this period and the diffusion of burial practices resulted in an unprecedented number of well-preserved human remains. This skeletal record, dovetailed with archeological, environmental, and chronological contexts, allows testing of hypotheses regarding biological processes at the population level. In this article, we review key studies about the biology of Upper Paleolithic populations based primarily on European samples, but integrating information from other areas of the Old World whenever possible. Data about cranial morphology, skeletal robusticity, stature, body proportions, health status, diet, physical activity, and genetics are evaluated in Late Pleistocene climatic and cultural contexts. Various lines of evidence delineate the Last Glacial Maximum (LGM) as a critical phase in the biological and cultural evolution of Upper Paleolithic populations. The LGM, a long phase of climatic deterioration culminating around 20,000 BP, had a profound impact on the environment, lifestyle, and behavior of human groups. Some of these effects are recorded in aspects of skeletal biology of these populations. Groups living before and after the LGM, Early Upper Paleolithic (EUP) and Late Upper Paleolithic (LUP), respectively, differ significantly in craniofacial dimensions, stature, robusticity, and body proportions. While paleopathological and stable isotope data suggest good health status throughout the Upper Paleolithic, some stress indicators point to a slight decline in quality of life in LUP populations. The intriguing and unexpected incidence of individuals affected by congenital disorders probably indicates selective burial practices for these abnormal individuals. While some of the changes observed can be explained through models of biocultural or environmental adaptation (e.g., decreased lower limb robusticity following decreased mobility; changes in body proportions along with climatic change), others are more difficult to explain. For instance, craniodental and upper limb robusticity show complex evolutionary patterns that do not always correspond to expectations. In addition, the marked decline in stature and the mosaic nature of change in body proportions still await clarifications. These issues, as well as systematic analysis of specific pathologies and possible relationships between genetic lineages, population movements and cultural complexes, should be among the goals of future research. Yrbk Phys Anthropol 51:70–99, 2008.

In these times, humans did not practice agriculture and Nature generously offered all her fruits

Plato

The Upper Paleolithic is generally associated with the emergence of modern humans and of modern patterns of behavior. While technology played a role in human adaptive strategies starting around 2.5 million years ago, the proliferation of technocomplexes after 40,000 BP points to the increased prevalence of culture as an environmental buffer (see Gamble et al., 2004). Not coincidentally, full-time colonization of northern latitudes also occured during the Upper Paleolithic (van Andel, 2003). Much of the discussion concerning this period has centered on issues of taxonomy and phylogeny. Beginning in the 1970s, however, growing attention has been paid to microevolutionary patterns within Upper Paleolithic populations themselves. The large number of ecozones, contrasting landscapes, and marked climatic fluctuations of the Last Glacial period provided challenges that affected patterns of gene flow, genetic drift, and biocultural adaptations. Moreover, population expansion over much wider territories made maintenance of proximity-based social networks more difficult (Gamble, 1986). The appearance of systematic use of decorated objects probably played an important role in establishing individual identity within increasingly complex social structures (Wobst, 1976) and in reinforcing social cohesion between groups in the absence of physical proximity (Gamble, 1999).

The unprecedented richness of the Upper Paleolithic archeological and fossil records offers excellent opportunities to test hypotheses about microevolutionary processes in these last Ice Age hunter–gatherers and to evaluate their adaptive strategies. Burials provide important information not only about the dead, but also about the living (Işcan and Kennedy, 1989; Pearson, 2003). While it is not possible to reconstruct the multitude of details that make up an individual life, the skeleton provides a record of events reflecting individual

B.M. Holt dedicates this paper to the memory of her mother, Janine Müller, who passed away before its completion.

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behavior and life conditions. Patterns of distribution of these behaviors, articulated in their proper archeological context, can yield important clues about the way past populations interacted with their cultural and physical environments.

Our goal in this article is to review key studies concerning issues of physical appearance, genetic relationships, health, diet, and physical activity of Upper Paleolithic populations, and to examine these issues in light of the important ecological and cultural developments that formed the adaptive context of these populations. We begin with a review of these developments.

BACKGROUND ISSUES

Definition of the Upper Paleolithic

A preliminary and central issue lies with the nature of the Upper Paleolithic (UP) as an economy, a technocultural complex, or a time period. In this section, we briefly review this problem in order to delineate carefully the UP sample considered in this study. Dates are reported uncalibrated. The Upper Paleolithic, a term coined in the context of Western European archeological tradition, refers historically to the period during which modern humans replaced Neandertals (Bar-Yosef, 2002). Much controversy surrounds the exact nature and timing of the transition between Middle and Upper Paleolithic (see for example Lindly and Clark, 1990; McBrearty and Brooks, 2000; Zilhão and d'Errico, 2003; Mellars, 2005), Some of the distinctive characteristics of the UP include the following: improved hunting technology, more diverse tool kits (including a preponderance of prismatic blades and tools made of bone, ivory, and antler), long-distance raw material trade networks, ubiquitous use of ochre and personal decorations (ivory and shell beads and pendants), systematic use of portable and parietal art (carved figurines and painted or engraved cave and rockshelter walls), and ceremonial burials. Some of these elements appear in Middle Paleolithic contexts. For example, blade technology and evidence of symbolic behavior are found in some Middle Paleolithic and Middle Stone Age assemblages (Henshilwood et al., 2002; Bar-Yosef, 2002). These previously sporadic occurrences became systematic components of assemblages only later, around 45 kya (thousands of years ago) (Bar-Yosef, 2002; Kuhn et al., 2001), suggesting significant behavioral shifts. For the purpose of this study, we will consider as UP any assemblage and associated human remains dated between 40 and 10 kya that include elements traditionally accepted as UP components (see preceding text). Although Châtelperronian and other Initial Upper Paleolithic industries (Bohunician, Szeletian, and Uluzzian) belong to the complex of UP cultures, because our focus is on modern humans, our analysis will not include associated Neandertal remains such as St. Césaire (Lévêque and Vandermeersch, 1981).

The issues that have emerged from the "out of Africa" debate and global characteristics of late Pleistocene climatic events suggest that a holistic perspective on UP adaptations must include information from Asia and Africa. Unfortunately, the paucity of EUP African and Asian skeletal material limits extensive comparisons. The European archeological and skeletal records are much richer and more complete, in part due to the long historical tradition of research there. For that reason, we focus primarily on European material, although comparisons will be made, when appropriate, with the largest African and Asian samples.

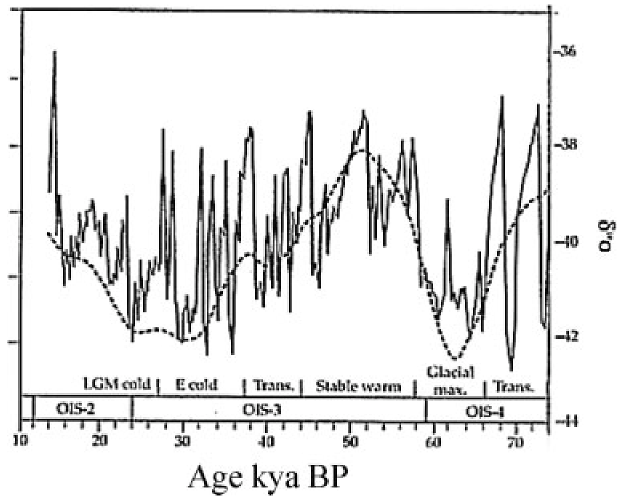

Fig. 1. Climatic fluctuations, based on the Greenland GISP2 δ^{18} O climatic record. Note the general downward trend after a brief warming period around 40 kya, and the rapidly spaced minor fluctuations. (Adapted from van Andel et al., 2003).

Environmental/climatic context

The UP emerged in Europe during the last Würm-Weichsel Interpleniglacial, around 40 kya, and spanned the second half of Oxygen Isotope Stage (OIS) 3 and all of OIS 2 (Fig. 1; Straus, 1995; van Andel et al., 2003; Mellars, 2006). During that time, a gradual decline in temperature resulted in the buildup of large ice sheets over most of northern Europe and the lowering of sea levels, a process that cumulated during the Last Glacial Maximum (LGM), around 20 kya, the time of lowest sea levels.

Following the relatively temperate first half of OIS 3, deterioration of climatic conditions began to accelerate (see Fig. 1). Evidence from ice and pollen cores summarized by the Stage 3 Project (van Andel and Davies, 2003) shows that the period between 42 and 34 kya saw rapidly spaced retreats and advances of the Fennoscandian ice sheet, resulting in high frequency climatic oscillations (van Andel, 2003), particularly in Europe. After the brief and warmer Hengelo event around 34 kya, conditions worsened again, eventually bottoming out around 20 kya. This event is recorded globally in ice cores showing sharp drops in sea levels (Lambeck and Chappell, 2001). Reduced rainfall, lower temperatures, and shorter growing seasons led to substantial reduction in environmental productivity in Europe and most of Africa and Asia (Gamble 1986; Brooks and Robertshaw, 1990; Close and Wendorf, 1990; Baruch, 1994; Bar-Yosef, 1996; Pookajorn, 1996; Kershaw et al., 2001; van Andel, 2003).

Temperatures began to rise steadily after 16 kya, as ice sheets retreated. Although this period saw alternations of relatively humid and warm conditions with cold and arid intervals (Straus 1995), these oscillations were not as deep as those preceding the LGM. One exception occurred around 11 kya (Younger Dryas or Dryas III), at which time glaciers expanded and global temperatures dropped sharply (Straus 1995). This was a relatively short lived, albeit severe, perturbation, and by 10 kya, sea levels and temperatures began to rise again, signaling the end of the Last Glacial and establishment of the Holocene Interglacial.

Although the overall pattern of climatic deterioration and subsequent amelioration is clear, the trends mask a considerable number of brief but marked climatic oscillations that contributed to environmental instability, undoubtedly impacting human population movements and strategies significantly (Davies et al., 2003; van Andel, 2003). In addition, the extremely varied topography of the European landscape, including several large west–east-oriented mountain ranges, resulted in numerous zones of refuge in which climatic impact would have been mitigated (Davies et al., 2003; van Andel, 2003). For instance, comparisons of pollen cores from northern Europe and Mediterranean regions show that warm intervals were fewer and shorter in the north and that recovery from cold intervals occurred much faster in the south (van Andel, 2003). In the Mediterranean region, glaciers were confined to the Pyrenees, Alps, and Apennines, resulting also in lower levels of precipitation and steppe conditions, but not as extreme as those found in central Europe. Southern Europe (peninsular Italy and Spain, the Balkans, and Greece) saw relatively moist and temperate climes, slightly cooler and drier in Spain and Portugal because of colder, dry winds sweeping from the east (Farrand, 1971).

Some groups occupied certain areas preferentially. Aurignacian sites, for example, tend to be located in mid-latitudinal and southern zones, while Gravettian sites are more numerous in northern plains, including areas above 50° N (Davies et al., 2003). Some areas, such as the Dordogne and Moravia, however, show continuous occupation, a pattern that Davies et al. attribute to relatively stable availability of desirable resources.

As mentioned in preceding text, the effects of the LGM were also felt in Africa and Asia. In some regions, however, the impact was mediated mainly by topography. In southwest Asia, for instance, while many areas saw much colder and drier conditions during the LGM, rainfall and forest cover remained stable in coastal hilly zones, a pattern that was reflected in population settlement and mobility (Bar-Yosef, 1996). For example, Kebaran settlements during that time were limited to coastal mountains. Following the LGM, regional precipitation increased gradually, resulting in rapid expansion of rich territories and population increase (Bar-Yosef, 1996).

A similar pattern obtained in North Africa, where the dry interior of the Maghreb remained unpopulated in favor of coastal regions from eastern Morocco to Tunisia (Close and Wendorf, 1990). In other areas of North Africa, the post-LGM period saw a rapid rise in rainfall and consequent contraction of the desert and development of new hunting-gathering territories (Close, 1996). In the lower Nile Valley, the annual flooding cycle fostered seasonal exploitation of rich resources even during the LGM (Close and Wendorf, 1990). Pollen profiles in most of tropical Africa reveal a period of arid conditions beginning around 20 kya in West Africa and 13 kya in East Africa (Brooks and Robertshaw, 1990). In South Africa, as elsewhere, temperatures increased after the LGM, but remained much colder than today until around 11.5 kya (Mitchell et al., 1996). In Asia, much drier conditions prevailed, with tropical forests remaining only in southern regions (Kennedy and Deraniyagala, 1989; Kershaw et al., 2001).

A brief survey of UP floral and faunal profiles serves to highlight some important contrasts. During the period leading to the LGM, combinations of aridity and extreme cold led to deforestation over much of Europe, resulting in open, dry steppe–tundra environments, rich enough to support herds of large game as well as carnivores, such as hyena, wolf, and bear (Gamble, 1986). Thus, while the cold temperatures and short growing seasons provided few plants for humans to exploit, the availability of large mammal herds afforded fairly reliable and relatively noncostly resources (Gamble, 1986).

The end of the LGM initiated a long period of fluctuating temperatures, retreating glaciers, rising sea levels, and reforestation, as evidenced by the marked increase in arboreal pollen after the LGM (Gamble, 1986). This process continued until 13 kya. The environment during this Tardiglacial period was still mostly one of steppes, with large herds of reindeer, bison, and horse, but the mammoth and wooly rhino that had thrived during the early part of the UP had disappeared. Then, following the Dryas III oscillation, the Late Glacial Interstadial (13–11 kya) saw further rising temperatures to quasi-postglacial levels, continuing reforestation (the Allerød pollen zone), and significant changes in faunal composition (Butzer, 1971; Gamble, 1986; Straus, 1995). One of the consequences of these climatic upheavals may have been an increase in foraging costs for human populations. Although the post-LGM period environments were rich in species such as roe and fallow deer, these species may have been more costly to exploit than open tundra and steppe fauna (Gamble, 1986).

Cultural context of the Upper Paleolithic

A comprehensive review of the remarkably complex UP archeological record is beyond the scope of this paper. Hence, we focus here on several aspects that can provide a cultural context within which to interpret biological evidence of human adaptive strategies during the Last Glacial. As previously mentioned, the UP is generally associated with major technological innovations. None of these innovations, however, appeared in systematic fashion until after the LGM (Straus, 1995). The spearthrower and harpoon, for instance, did not appear until after the LGM (Schmitt et al., 2003). Although the recently discovered Grotte Chauvet (Ardèche, France) has been dated to around 31 kya (Valladas et al., 2001), parietal art, another UP hallmark, did not gain in importance until later (Straus, 1995). Similarly, hunting of large herds of horse and bison appeared in greater frequency only after 20 kya (Straus, 1987, 1995; Enloe, 1993; Pike-Tay, 1993). The appearance of these technologies and of distinct organizational systems following the LGM form the basis for the traditional division of the UP into two distinct phases, Early Upper Paleolithic (EUP), from 40 to 20 kya, and Late Upper Paleolithic (LUP), from 19 kya to the end of the Last Glacial, around 10 kya (see Fig. 2).

The Aurignacian represents the oldest UP tool tradition in Europe. In the absence of Aurignacian burials in Europe, its association with anatomically modern humans is still a matter of great debate (d'Errico et al., 1998; Churchill and Smith, 2000; Trinkaus, 2005), but there is wide agreement that the Aurignacian includes elements that are generally associated with modern human behavior (Bar-Yosef, 2002). Although its roots may lie outside Europe, perhaps in the Near East

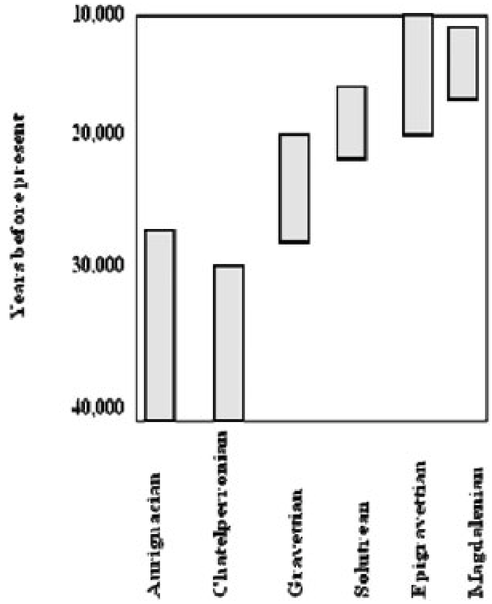

Fig. 2. Uncalibrated chronology of Upper Paleolithic cultural periods in Europe.

Fig. 3. Ivory carving of a horse from the Aurignacian site of Vogelherd (Germany). (Courtesy of University of Tübingen and N. Conard). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

Fig. 4. Details of Arene Candide 1 (Il Principe) (Italy), one of the most spectacular examples of Gravettian ceremonial burials. (Courtesy of the Soprintendenza Archeologica della Liguria and O. Levati). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

Fig. 5. Burial of a child (AC 8) from the Epigravettian necropolis from Arene Candide (Italy). Abundance of ochre and rich ornamentation (note a number of squirrel caudal vertebrae on the thorax) recall Gravettian burial pattern. (Courtesy of the Soprintendenza Archeologica della Liguria and O. Levati). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

(Bar-Yosef, 2002; Kuhn et al., 2001), the Aurignacian first appeared in Western Europe at the end of the Interpleneglacial, around 37 kya and disappeared by 28 kya. Characterized by split-base points, retouched blades, nosed end-scrapers, and the ubiquitous presence of ornaments (see Fig. 3), Aurignacian assemblages are extremely variable. Contemporaneous and/or earlier regional industries include the Châtelperronian (France/Spain), Bohunician (Moravia), Szeletian (central Europe), and Uluzzian (Italy).

The next EUP technocomplex, the Gravettian, appeared around 28 kya, in the context of a sharp climatic degradation. The origin and dispersal of the Gravettian are still debated, with central and/or eastern Europe as strong candidates (Otte, 1990; Svoboda et al., 1996), while genetic evidence points to population movements from the Near East at the time of appearance of this culture (Richards et al., 2000). Gravettian assemblages eventually replaced Aurignacian ones, although recent calibrated dates suggest a long period of overlap between the two technocomplexes (van Andel, 2003). Characterized by backed (Gravette) blades retouched into sharp points, Gravettian tool kits included novelties such as tanged and shouldered points as well as mammoth ivory objects (White, 2003). Mobile art and ornamentation were often found associated with burials (see Fig. 4), and multiple burials became frequent (Formicola, 2007). Although EUP archeological assemblages generally indicate much higher mobility than those of the Middle Paleolithic (Bar-Yosef, 2002), this is particularly true of the Gravettian (Hahn, 1987; Scheer, 1993), prompting Gamble to refer to the EUP as an "open system" (Gamble, 1986). Studies of lithic procurement patterns show that Gravettian groups often traveled distances in excess of 200 km to obtain materials necessary for tool production (Kozlowski, 1991; Svoboda et al., 1996; Flébot-Augustin, 1997; Negrino and Starnini, 2003). In contrast with Mousterian and Aurignacian groups, who appeared to have retreated southward in periods of severe climatic deterioration, the number of Gravettian sites in northern and central Europe continued to rise during harsh conditions of the early LGM (Abramova et al., 2001; Verpoorte, 2002; van Andel, 2003). Some northern areas, in fact, show signs of reoccupation prior to the end of the LGM (Otte, 1990), while others remained continuously occupied despite the deteriorating climatic situation (Montet-White, 1994; Terberger and Street, 2002). Organizational systems included complex social long-distance networks designed to cushion the impact of a highly seasonal environment and to ensure maintenance of social cohesion, mating networks, and exchange of information between small demes distributed across a vast territory (Gamble, 1986; Gamble et al., 2004; see also Whallon, 2006). The extensive use of portable art, such as beads and the so-called Venus figurines, probably functioned to extend social connections in the absence of proximity (Gamble et al., 2004). The striking resemblance in shape and size of these figurines and their geographic distribution across 2000 km during that period stress the maintenance of a common system of beliefs among distant populations (Gamble, 1986; Mussi et al., 2000).

LUP assemblages point to the breakup of this homogeneity and to rising regional diversification (Rozoy, 1989; Bosinsky, 1990), a process reflected in art, lithic technology, and possibly in funerary behavior. While the Gravettian model of inhumation continued in areas of diffusion of the Epigravettian (Fig. 5; Mussi, 2001), in central Europe, funerary practices that included defleshing, dismembering, and differential treatment of the skull appeared during the Magdalenian (Le Mort and Gambier, 1991; Gambier and Le Mort, 1996; Orschiedt, 2002). Factors underlying the rise in regional heterogeneity included increased population density, decreased mobility, and intensified exploitation of available resources. Reduced rainfall, lower temperatures, and shorter growing seasons associated with the LGM led to a decline in productivity (Dennel, 1983; Gamble, 1986; Holt et al., 2000). Demographic stress resulted from reduction in available territory and increased population density (Mellars, 1985; Clark and Straus, 1986; Jochim, 1987; Mussi and Zampetti, 1988), coupled with decreased resource reliability (Straus, 1995).

Appearance of the Solutrean technocomplex probably resulted from the double stress imposed by rigorous climatic conditions and increased population density. As a result of increasingly severe environmental degradation, northern Europe was mostly abandoned. The rise in the number of LUP sites in the more temperate area of southwest Europe points to an increase in population density (Mellars, 1985; Jochim, 1987; Mussi and Zampetti, 1988). Cachel (1997) suggests that the crucial need to maintain an optimum protein/dietary fat ratio may have also contributed to the population increase and decreased mobility[1], as groups aggregated around sources of dietary fat, such as anadromous fish.

Several lines of evidence indicate a significant shift in resource exploitation strategies during the LUP, and the Solutrean in particular. The development and refinement of throwing technology, such as points that were probably used as projectiles thrown by the newly introduced spearthrower (atl-atl), indicate diversification of subsistence (Freeman, 1973; Straus, 1990, 1993). Diversification of the food base to include fish, shellfish, and birds suggests a need to cope with demographic stress by adding low-yield/high-cost foods (Straus et al., 1981; Straus, 1986; Stiner, 2001; Stiner et al., 1999). While uneven distribution of resources generally results in higher foraging mobility (Winterhalder, 1981; Keeley, 1988), reduced availability of exploitable territory forced groups to intensify exploitation of existing resources (Fig. 6; Straus and Clark, 1986; Stiner et al., 1999). This pattern of diet breadth expansion is recorded in non-European LUP sites as well (Nadel and Hershkovitz, 1991; Mitchell et al., 1996). Finally, LUP site structure and associated fauna suggest a rise in logistical exploitation of resources (Straus, 1986, 1990; Enloe, 1993; Pike-Tay, 1993; Svoboda et al., 1996), as indicated by increased use of specialized camps in Solutrean deposits (Straus, 1986, 1990).

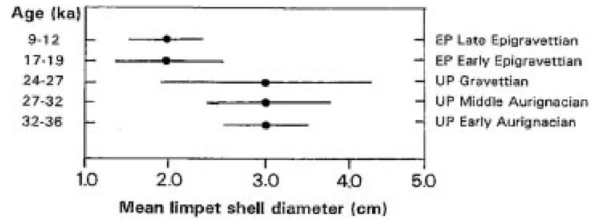

Fig. 6. Trend in reduction of limpet size from the site of Riparo Mochi (Italy). Note the sharp decrease in size between EUP (Gravettian) and LUP (Epigravettian). (From Stiner M, et al., 1999. Paleolithic population growth pulses evidenced by small animal exploitation. *Science* 283:190–194. Reprinted with permission from the American Association for the Advancement of Science).

Beginning around 16 kya, improvement of climatic conditions triggered the reexpansion of human populations northward and major changes in the LUP fauna. Although reindeer, horse, and bison remained, archaic Pleistocene fauna such as the mammoth disappeared, probably the result of reduced steppe–tundra habitats. As suddenly as they appeared, Solutrean industries disappeared around 16 kya, to be replaced by increasingly microlithic points and bladelets fixed in antlers, as well as wood points and geometric shapes (Jochim, 2002). In western Europe, the Magdalenian replaced the Solutrean, while in central and southeast Europe, there was continuity between Gravettian and Epigravettian assemblages (Bietti, 1990; Mussi, 2001). Appearance of large-scale specialized hunting of target species, documented at sites such as Pincevent and Duruthy in France, and Petersfelt in Germany (Straus, 1995) points to increased refinement of systematic and efficient logistic hunting strategies during the Magdalenian.

The presence, in southwest France, of relatively sedentary, complex hunter–gatherer groups during the later part of the LUP (Mellars, 1985; Jochim, 1987) attests to further reduction in mobility. In addition, evidence of site modification, such as pavement and wall construction indicating relatively constant use of habitation area, of burial proliferation, and of development of parietal art, suggest increased attachment of local groups to sites, and increased territoriality, perhaps linked to greater intergroup differentiation (Boquet Appel and Demars, 2000; Jochim, 2002). According to this view, the development of parietal art as ethnic markers reflects increased intergroup differentiation in Europe during the LUP (Jochim, 1987).

[1]Mobility refers here to the foraging distance covered daily by an individual.

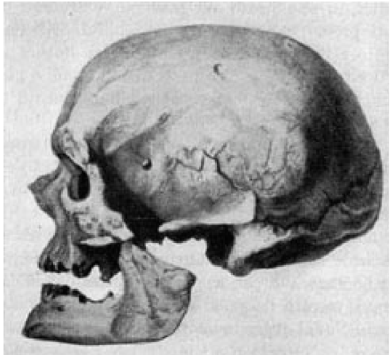

Fig. 7. Cro-Magnon 1 (lateral view). This skull was discovered during the 19th century in the rockshelter of Cro-Magnon, Dordogne (France). Recognized right away as belonging to modern man, Cro-Magnon has become a symbol of Upper Paleolithic populations. (From Biasutti, 1967).

Fig. 8. Herto, lateral view (Middle Awash, Ethiopia). This cranium, radiometrically dated to around 160,000 BP, exhibits morphology intermediate between archaic African *Homo sapiens* and later anatomically modern humans. (From White TD, et al., 2003. Pleistocence *Homo sapiens* from Middle Awash, Ethiopia. *Nature* 423:742–747. Reprinted by permission from MacMillan Publishers Ltd.

THE SAMPLE

The studies that we included in our review of UP skeletal biology are based on modern human specimens. Because our goal was to focus on biocultural adaptations of UP populations, and not on the origins of modern humans, we have generally excluded specimens exhibiting "archaic" morphology attributable to Middle Paleolithic groups, such as the Neandertals. To the extent that they are relevant to the discussion regarding UP craniodental morphology, we discuss some EUP modern human cranial materials that exhibit morphology that suggests possible admixture, such as crania from Mladeč (Frayer et al., 2006; Wolpoff et al., 2006) and Peştera cu Oase (Trinkaus et al., 2003). We focus primarily on European skeletal samples, but include a discussion of non-European material whenever sample sizes or preservation allow meaningful comparisons. Our goal in this review is to present a thorough survey of the latest studies regarding the biology of UP populations and, in particular, to articulate biological data within environmental and archeological contexts. In addition, however, we feel that such a review should also serve as a reference tool. Therefore, we have included a list (see Appendix) of UP skeletal material from Europe, Asia, and Africa. This list is not exhaustive and includes only material comprising at least a cranium and/or relatively well preserved postcranial elements. Specimens are grouped into EUP and LUP samples, in keeping with the scheme used throughout the paper.

THE BIOLOGY OF UP PEOPLE

Craniodental morphology

UP anatomically modern humans differ from "archaic" populations in having taller and anteroposteriorly reduced (shorter) cranial vaults showing parietal expansion and clear bossing, more vertical foreheads, orthognatic, "tucked under" faces, coronally oriented zygomatics, a canine fossa, lack of distinct superorbital tori, reduced nasal aperture, smaller anterior teeth, and a projecting chin (see Fig. 7). Some of these characteristics emerge in a mosaic fashion, probably as early as 200 kya (Fig. 8; White et al., 2003; McDougall et al., 2005). The evolution of the modern human morphology lies at the center of intense debate regarding the relationship between UP populations and the Neandertals (Smith et al., 1989; Bräuer, 1990; Lahr, 1994; Holliday, 1997; Wolpoff et al., 2001; Stringer, 2002). While a consensus has emerged that modern humans appeared in Africa at least by 200 kya and spread subsequently to Europe and Asia, much discussion remains about the amount of gene flow that took place between local populations and the newcomers (Stringer, 2002; Trinkaus, 2007).

Presumed manifestations of Neandertal traits in UP crania, such as occipital bunning and suprainiac fossa, are alternatively interpreted as plesiomorphic retention (Stringer and Andrews, 1988) or evidence of hybridization (Stringer, 2002; Garralda, 2006; Trinkaus, 2007). The lack of reliably dated human remains associated with Aurignacian assemblages has further complicated the issue (Churchill and Smith, 2000). To make matters worse, accelerated mass spectrometry (AMS) ^{14}C redating of fossil crania has failed to confirm the Aurignacian status of some of these specimens, further reducing the list of candidates for "makers of the Aurignacian" (Conard et al., 2004; Trinkaus, 2005; Soficaru et al., 2006). The largest sample of reliably dated modern human remains comes from Mladeč (Czech Republic) (Wild et al., 2005; Teschler-Nicola, 2006 and references therein). This 31-kya-old sample exhibits diagnostically modern human features (Churchill and Smith, 2000; Trinkaus, 2005; Frayer et al., 2006; Wolpoff et al., 2006). The presence of Neandertal traits in some of the crania, however, such as a suprainiac fossa in Mladeč 6, midfacial prognathism in Mladeč 1 and 2, and occipital bunning in Mladeč males in general (see Fig. 9), has been taken by some as evidence of Neandertal contributions to the modern human gene pool (Churchill and Smith, 2000). The recently discovered remains from Peştera cu Oase (Romania), dated at 35 kya, also show a mix of modern and archaic features (Trinkaus et al., 2003; Rougier et al., 2007). While Oase exhibits a clearly modern cranial and mandibular morphology, Trinkaus et al. point to the flat frontal profile, wide mandibular ramus and large distal molars as further evidence of regional continuity.

The evolution of modern cranial shape (i.e., increased cranial height, expanded parietals and tucked under face), follows a different trajectory than that of cranial size and robusticity, robusticity referring here to the muscle scarring and the degree of expression of superstructures such as supraorbital and occipital ridges

(Lahr, 1996; Lahr and Wright, 1996; Churchill, 1998). While North African and Levantine late Pleistocene crania such as Herto, Omo 1, and those from the Qafzeh and Skuhl caves are clearly modern in shape, they are notably more robust than most UP moderns, i.e., exhibit larger superstructures (Kidder et al., 1992; Van Vark et al., 1992; Lahr and Wright, 1996). Various studies have pointed out a general reduction in craniodental size during the UP (Frayer, 1978, 1984; Schumann, 1995), particularly after the LGM.

UP reduction in craniodental robusticity is said to parallel dietary changes and improving technology that resulted in lower levels of craniofacial strains (Brace, 1962; Frayer, 1978, 1984). This mechanical model relates facial robusticity, such as supraorbital ridge formation, to the amount of bending stress engendered by anterior dental loading (Endo, 1966; Russell, 1985; Demes and Creel, 1988). Relaxed selection for large anterior tooth size resulting from increased reliance on technology combines with lower levels of musculoskeletal stress to produce the gracile morphology typical of modern human crania. Frayer's (1978) seminal work on Late Pleistocene and Early Holocene patterns of dental evolution showed that most of the reduction in anterior tooth and palate size occurred between EUP and LUP groups, a change he relates to improving technology. This conclusion is supported by the proliferation of specialized technologies during the LUP (Straus, 1995). Similar patterns of craniodental gracilization are found in Asia (Kennedy et al., 1987) and the Levant (Hershkovitz et al., 1995).

The mechanical stress model has not gone unchallenged. Studies by Hylander and coworkers (Picq and Hylander, 1989; Hylander et al., 1991), for instance, have failed to confirm the link between bite force and glabellar masticatory strains. The authors find that, although the glabellar region does bend in response to bite force, strain magnitude in that region is too low to explain the formation of large supraorbital ridges, and is higher in other areas of the face. They argue that, rather than countering masticatory stresses, the supraorbital ridges fill a structural void between neurocranium and face.

Other interpretations of craniofacial gracility come from studies of developmental correlation between cranial dimensions and robusticity (Lahr and Wright, 1996; Lieberman et al., 2000). Lahr and Wright (1996) found a positive correlation between cranial length and breadth and expression of superstructures in a large sample of recent and fossil modern humans. In particular, they showed that longer crania were significantly more likely to have robust brow ridges and zygomatic trigones. The same association, although not as strong, was found between narrow skulls and large teeth and large superstructures. Studies of the correlations between occipital projection or "bunning" and cranial dimensions (Lieberman et al., 2000; Gunz and Harvati, 2007) suggest that the presence of occipital buns in central European early modern human crania may reflect size-related structural integration. Polanski and Franciscus (2006), on the other hand, found little integration between neurocranium and face in the modern human cranium.

Clearly much work remains to be done in the area of cranial developmental morphology, but the evidence of structural integration points to the importance of understanding character intercorrelation when using these traits to distinguish the effects of competing models of selection, gene flow, and drift on cranial variation.

Given that the major organizational shifts and population contraction at the onset of the LGM undoubtedly disrupted the large mating networks of the EUP (Gamble, 1986), it is reasonable to assume that increased regionalization ensuing from reduced gene flow between populations should be detectable in the form of higher interpopulation variability in genetically based phenotypic traits, such as cranial size and shape. Frayer (1988) proposed such a change between UP and Mesolithic, arguing that regionalization linked to increased sedentism in early Holocene should lead to accumulation of genetic variability among Mesolithic populations. Using 21 cranial metric variables, he found much higher differentiation within Mesolithic than UP samples, confirming a rise in regionalization.

As first suggested by von Bonin (1935), UP groups exhibit little variability. Gambier (1989) also found high levels of similarities in cranial shape within the EUP in her comparison of western and central European groups, concluding that the main differences lie with the robustness of central European crania. Her study, however, included only Aurignacian and Gravettian crania, and so did not address the issue of regional differences in the LUP. Based on these studies, European UP cranial morphology appears to have been relatively homogeneous when compared with subsequent Mesolithic populations. It is possible that selection for robust crania during the UP constrained morphological variability, trumping the effects of fluctuating gene flow. As selection relaxed with increased reliance on cultural adaptation (see discussion in preceding text), the effect of genetic drift within early Holocene local populations resulted in a rise in interpopulation variability.

Thus, despite evidence that patterns of decreased mobility and increased territoriality emerged by the onset of the LUP, regionalization does not appear to have resulted in detectable levels of interpopulation genetic variability, at least based on cranial morphology.

Body size and shape

Stature represents an important component of body size and a distinguishing characteristic at both individual and population levels. Like most polygenic traits, phenotypic expression of stature depends on the complex interaction of genes and environmental factors (Bodmer and Cavalli Sforza, 1976; Susanne, 1985; Eveleth and Tanner, 1990). Consequently, stature lends itself well to inferences about socioeconomic conditions in past populations (Larsen, 1997).

From the discovery of the first UP remains from Cro-Magnon and Grimaldi, the high stature of these early populations has been evident (see Fig. 10). Quantifying this variable, however, has been more difficult. Rivière (1887) estimated the height of the individuals from Bausso da Ture and Caviglione (Grimaldi) to be around 200 cm, while Verneau (1906), using Manouvrier's (1892) tables, suggested a value of 187 cm for the whole Grimaldi sample. More reliable regression equations have greatly reduced these early estimates, although they are still remarkably high. The accuracy of the estimate depends on the similarity between body proportions of the reference population used to derive the equation and those of the sample under study. Comparisons of estimates provided by different limb bones (Parenti, 1971; Formicola, 1983) and results based on Fully's (1956) anatomical method (Formicola, 1993, 2003) may help clarify

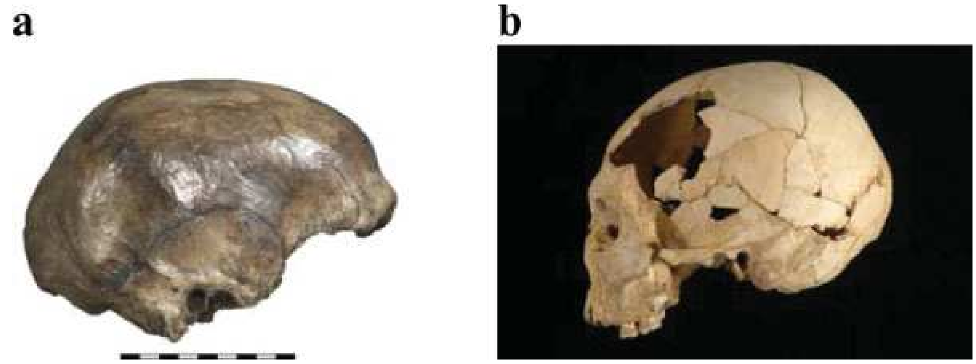

Fig. 9. (a) Mladeč 5 (lateral view), (b) Peştera cu Oase 2 (lateral view). These two crania are particularly interesting for their antiquity, radiometrically dated at more than 30,000 years BP and for the mosaic pattern of modern and archaic traits they exhibit. (Mladeč 5: Courtesy of Naturhistorisches Museum of Vienna and M. Teschler-Nicola; Oase 2: From Trinkaus E, et al., 2003. Early modern human cranial remains from the Peştera cu Oase, Romania. *Journal of Human Evolution* 45:245–253. Reprinted with permission of Elsevier © 2003.). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

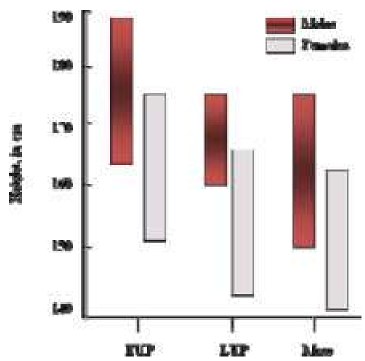

Fig. 11. Stature variability in Early (EUP), Late (LUP) Paleolithic and Mesolithic (Meso) European samples. The differences between EUP and LUP are highly significant in both males and females. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

the issue. Stature variations shown in Figure 11 are taken from Formicola and Giannecchini (1999) and are based on equations provided by Formicola and Franceschi (1996). These equations are derived using the major axis regression method that allows extrapolation outside the range of variation of the sample and are therefore particularly applicable to situations in which large deviations from the mean can be expected (Feldesman and Lundy, 1988). Independent of the choice of equations, however, one obvious trend characteristic of the UP is the sharp reduction in stature between EUP and LUP for both males and females. EUP European males and females are taller and heavier than their LUP counterparts (Fig. 11, Table 1). This trend continues into the Mesolithic in western Europe, although the changes are much less accentuated. As with craniodental morphology, therefore, the LGM appears to have been a critical period as first pointed out by Jacobs (1985).

Reduction in gene flow and decline in nutritional conditions have been suggested as the main factors

Fig. 10. Grotte des Enfants 4 (Grimaldi, Italy) clearly exhibits the characteristic linear body proportions of Gravettian populations. (Courtesy of the Musée d'Anthropologie Préhistorique de Monaco and P. Simon). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

TABLE 1. Stature and body mass differences, by time period, sex, and region

	EUP[a]	LUP[a] (Europe)	LUP[b] (Southeast Asia)	LUP[b] (Nile valley)	LUP[b] (Mediter.)
Males					
Stature (cm)	174.1[c] (1.5, 10)	165.3 (3.7, 15)	153.1 (5.6, 3)	170.2 (1.2, 15)	168.4 (1.3, 12)
Body mass (kg)	71.0 (1.7, 10)	67.9 (2.7, 15)	56.8 (5.9, 4)	64.1 (1.4, 21)	66.2 (1.7, 15)
Females					
Stature (cm)	161.8 (3.2, 5)	154.5 (3.4, 7)	147.8 (2.17, 7)	162.4 (0.7, 16)	161.7 (2.4, 4)
Body mass (kg)	60.7 (2.4, 5)	55.5 (3.5, 7)	52.7 (1.5, 8)	59.7 (0.9, 21)	58.3 (4.0, 5)

[a] Data from Holt (2003).
[b] Data from Laura Shackelford (personal communication and Shackelford, 2007).
[c] Mean, with standard error and sample size in parentheses.

underlying the decrease in height between EUP and LUP (Formicola and Giannecchini, 1999). Increased LUP territorialism and restricted mating networks would have resulted in reduced gene flow. The role of gene flow in the positive trend in stature in modern populations has been suggested by Malina (1979) and Little and Malina (1986). Given, however, that breakage of the isolates and consequent exogamy in recent populations occur concomitantly with improved health and nutritional conditions, the importance of genetic versus socioeconomic factors remains obscure (Susanne, 1985).

Observations on recent populations have repeatedly stressed the importance of life condition variables such as nutrition as a main factor in the growth process (Malina, 1987; Steckel, 1995). As discussed in preceding text, archeological evidence shows that climatic degradation during the LGM caused an increase in population density (Straus, 1995) and a pattern of resource overexploitation (Clark and Straus, 1986; Stiner et al., 1999). Expansion of diet breadth to include low-rank resources such as aquatic resources is supported by isotope analysis (Richards et al., 2005; see following text). While these data suggest that a decline in availability of protein might have played a role, neither the faunal evidence nor biological stress markers and isotope data point to dramatic deterioration of life conditions and quality of diet (Brennan, 1991; see following text). It is possible, however, that in the context of increasingly difficult access to resources, there might have been selection for reduced body size in order to reduce metabolic and nutritional demands (Formicola and Holt, 2007).

Comparisons of European data with samples from western Asia (Mediterranean), Africa (Nile Valley), and Southeast Asia show that Mediterranean and Nile valley LUP groups are taller than LUP Europeans, while Southeast Asians exhibit particularly low stature (Table 1; Shackelford, 2007). Some Epipaleolithic Natufian individuals, in fact, are as tall as EUP Europeans (Nadel and Hershkovitz, 1991; Stock et al., 2005). Patterns of body mass are less clear. While male body mass decreases along a north–south cline, with the Mediterranean sample intermediate between European and Nile Valley groups, females exhibit an opposite trend.

Like cranial shape, body shape is to a large extent genetically determined (Eveleth and Tanner, 1990; Y'Edynak, 1978). This and the fact that body proportions are highly correlated with climatic factors (Roberts, 1953; Ruff, 1994) means that body shape can give clues about population affinities and climatic adaptation. In a comparison of Neandertal and EUP body proportions with those of extant high- and low-latitude human populations, Holliday (1995, 1997) showed that, in contrast with Neandertals, EUP groups have long legs and narrow trunks, suggesting that these early modern humans did not evolve in glacial environments. The body proportions of Gravettian populations are all the more surprising since, as discussed in preceding text, climatic data and chronospatial distribution of archeological sites show that, in contrast with Aurignacian and Mousterian groups, Gravettian adaptive strategies allowed them to continue occupying northern territories until about 22 kya, only retreating southward shortly before the LGM (van Andel, 2003). This underscores the improvements in organizational systems characteristic of the Gravettian. Considering a range of 28–20 kya for the Gravettian, the persistence of these "tropical" body proportions points to the conservative nature of those characters. Only in the LUP do body proportions approximate those of extant populations adapted to temperate zones. These changes include increased trunk breadth and trunk height to limb length ratio. Surprisingly, however, there is no change in brachial and crural indices (Holliday, 1999), suggesting that environmental factors other than climate may have acted on body proportions.

Some researchers (Wolpoff, 1989; Caspari, 1992) have argued, for instance, that the long legs of EUP populations represent an adaptation to high mobility levels. Given archeological evidence of high levels of EUP mobility, especially during the Gravettian (Gamble, 1986; Hahn, 1987), high energy costs associated with locomotion might have been an important selective factor (Weaver and Steudel-Numbers, 2005). As levels of mobility and associated energetic costs decreased during the LUP, selection for long legs would have been relaxed and climatic selective pressure may have become the dominant force shaping body proportions (Weaver and Steudel-Numbers, 2005). A survey of limb length and mobility levels among groups of extant foragers, however, did not find any correlation between long legs and high mobility (Holliday and Falsetti, 1995).

Adequate quality of nutrition has also been implicated as a determinant of body proportions (Bogin and Rios, 2003). Comparisons of the sitting height-to-leg ratio in a sample of Guatemalan Maya and American Guatemalan children show that the latter exhibit significantly longer legs relative to trunk height, a difference Bogin and Rios attribute to improved health and nutritional status. Archeological evidence of resource stress (Stiner et al., 1999) and of increased frequency of biological stress markers (see following text) in the LUP suggests a possible decline in life and nutritional conditions. It is important to point out, however, that differences in nutritional status during the UP in no way approximate those seen in more recent prehistoric times (Brennan, 1991). It is, therefore, unlikely that nutrition played a role in the change in body proportions.

Genetic relationships

Founder analysis has been used extensively in the past 10 years to trace population movements. The method consists of analyzing haplotype lineages from nonrecombinant DNA sequences (Y chromosome and mitochondrial DNA) in putative descendent and source populations, in order to identify and date population migrations (Richards et al., 2000). Founder analysis has figured prominently in the acrimonious debates on the respective contributions of Paleolithic and Neolithic population movements to modern European genetic variability. An in-depth review of this literature is beyond the scope and goals of this paper. However, given the marked effect the LGM had on shaping UP adaptive strategies, it is reasonable to expect that this impact should be reflected in some ways in the genetic structure of descendant populations.

Results from several studies of Y chromosome and mitochondrial (mtDNA) haplotype distribution in modern Europeans indicate that several lineages have roots in two separate Paleolithic founding events. Comparison of mtDNA haplotype cluster distribution by Richards et al. (2000) suggests that many of these clusters originated in the Near East, some as early as 50 kya. For most of the clusters surveyed, more recent, Neolithic, contributions rarely exceed 50%, contradicting the hypothesis of extreme Neolithic demic replacement based on biparental genetic systems (Chikhi et al., 1998). This pattern was confirmed and refined by more recent data based on complete genomes (for a review see Torroni et al., 2006). The basal branches of European mtDNA lineages, representing as much as 75% of present lineages, would be haplogroups R0 (previously known as pre-HV), JT, and U, whose variation can be dated to 40–50 kya. The age estimates and spatial distributions of R0 subhaplogroups H1, H3, V, and U5b strongly support recolonization of Europe from the Near East and Iberian LUP refuges approximately 15 kya (Pereira et al., 2005; Roostalu et al., 2007). These data correlate well with archeological evidence that most areas in northern Europe were abandoned in favor of southern refuges, such as the Iberian Peninsula and the Balkans, by at least 20 kya. Subsequent climatic amelioration saw a major wave of northward demographic expansions.

Evidence from Y-chromosome haplotypes paints a similar picture (Semino et al., 2000). Distribution of 22 markers in 25 European and Middle East geographic regions suggested that two major lineages, defined by derived M173 and M170 mutations, have roots in the Paleolithic. The M173 lineage (haplogroup R1 according to the most recent nomenclature; International Society of Genetic Genealogy, 2007) occurs with great frequency in European, Native American and some Siberian groups, a pattern that Semino et al. argue reflects an ancient Eurasian origin, possibly brought into Europe by an Aurignacian diffusion around 40 kya. The M170 lineage (haplogroup I), estimated to date to around 22 kya, probably descends from a basal haplogroup present in Middle East populations, perhaps the result of a second migration into Europe 25 kya. M170 subclades (I1a, I1b, I1c) show frequency and variation patterns that suggest independent population expansions during and immediately after the ice age (Rootsi et al., 2004). The differentiation within mtDNA haplogroup H and Y haplogroup I is estimated to have occurred around 25 kya.

The wide confidence margins associated with the dates for these genetic markers make connections to archeologically defined groups a risky venture (Gibbons, 2000). While much work clearly remains to be done to confirm the genetic evidence, these independent haplotype lineages lend support to the hypothesis of an exogenous origin for the Gravettian, a possibility raised by distinctive differences between Aurignacian and Gravettian assemblages (Jochim, 2002). In that context, it is possible that the mixed tropical body proportions of Gravettian populations (see preceding text) reflect an “out of Africa” migration, but a more recent one than that proposed by Holliday (1997). Analysis by Caramelli et al. (2003) of mtDNA sequences extracted from two Gravettian-age human skeletons from Italy, Paglicci 25 and Paglicci 12, dated to around 24 kya, provides additional support for a possible Near Eastern origin of the Gravettian. The presence, in both individuals, of haplogroups found quite frequently primarily in Near Eastern populations but rarely or much less frequently in Europeans, might reflect the same migration suggested by the modern distribution of haplotypes from mt-H and Y-I lineages (Torroni et al., 1998; Semino et al., 2000). On the other hand, correlation of calibrated radiocarbon dates and climatic evidence (van Andel, 2003) suggest that, unlike Mousterian and Aurignacian groups, the Gravettians had developed adaptive strategies that allowed them to continue occupying northern areas up to the LGM, in some cases reoccupying these zones well before temperatures started rising again (Otte, 1990). This would suggest a long, in situ, development and refinement of Gravettian adaptations, rather than a migration from temperate zones.

Finally, one cluster of the mtDNA haplogroups analyzed by Richards et al. (2000) and dated to the Middle Upper Paleolithic (MUP) occurs most frequently in Italy. This is particularly interesting, given strong evidence of cultural continuity in that region between Gravettian and Epigravettian cultures (Bietti, 1990; Mussi, 2001).

Paleopathology

Paleopathological literature regarding Paleolithic populations is extremely scattered, and there are few systematic studies. For this reason, our survey centers mostly on Europe. Although limited to remains from southwestern France, Brennan’s 1991 doctoral dissertation still represents the most frequently quoted report in analyses of Middle Paleolithic and UP health status. An additional important contribution at a regional level has recently been provided by Trinkaus and coworkers (2005) on the Gravettian material from Moravia. Our analysis begins with these data and integrates them with information collected from a plethora of papers focused mainly on anatomical descriptions.

Data collected on the Gravettian material from Moravia (Dolní Vestoniče and Pavlov) from Trinkaus et al. (2005) point to cranial trauma as a recurring pathology. Such lesions are generally of minor importance, except in two severely affected cases, DV 3 (trauma to the left side of the face, resulting in asymmetry and loss of condyle) and DV 11/12 (depressed frontal fracture). Limb involvement is less frequent, but clear evidence in that sample is provided by the healed ulna fracture exhibited

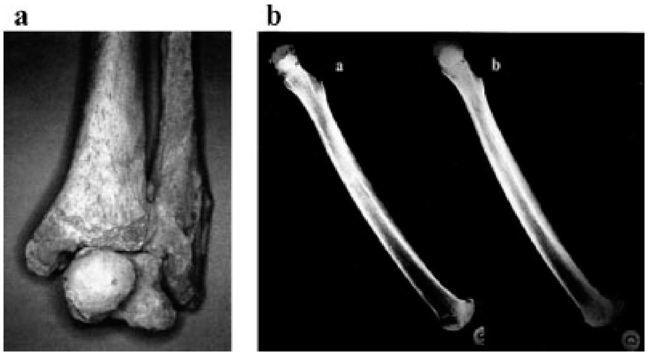

Fig. 12. (a) Detail of left ankle fracture in Vado all' Arancio (Italy). Note the extensive remodeling. (b) Radiographs of Vado all' Arancio's femora (lateral view). The right (normal) side shows hypertrophy of cortical bone.

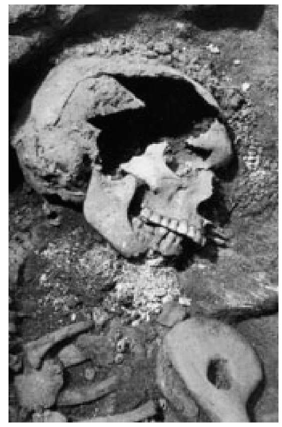

Fig. 13. Arene Candide 1 (Il Principe), at the time of discovery. Note the large lump of yellow ochre found under the mandible. (Courtesy of the Soprintendenza Archeologica della Liguria).

by DV 15. Additional traumatic lesions in the Gravettian include the healed radial fracture of the Caviglione 1 specimen (Verneau, 1906), a localized blunt trauma exhibited by the Mladeč 25c ulna (Teschler-Nicola et al., 2006), and, in more recent samples, the tibial fracture of Veyrier 1 (Pittard and Sauter, 1945) and the ankle fracture suffered by Vado all'Arancio 1 (Fig. 12; Holt et al., 2002). Brennan (1991) discusses LUP cases of healed cranial depressed fractures (Chancelade, Le Placard) and mentions two additional Magdalenian specimens (Laugerie Basse and Sorde) examined by Courville (1967). According to Courville, Cro-Magnon 2 is also affected by cranial trauma. Dastugue (1967) rejects this diagnosis based on the appearance of the lesions that is quite different from that caused by peri-mortem trauma, such as sharp edges, and the absence of radial fracture lines. Among Aurignacian remains, multiple traumatic but minor lesions have been diagnosed on the Mladeč 5 calotte (Teschler-Nicola et al., 2006). Regarding causes of these traumas, with the exception of the fracture affecting the ulna of DV 15 that could represent a parry fracture, there is no evidence of deliberate injury. Different scenarios are evoked by flint flakes, likely from javelins or arrows, embedded into the bones of two Late Epigravettian remains from Italy (Grotte des Enfants 2 and San Teodoro 4). In the latter case, a flint flake was found in the hip bone of an adult individual who survived the wound (Bachechi et al., 1997). In the case of the child from Grotte des Enfants, the flint was embedded in the T4 vertebral body (Henry-Gambier, 2001). Absence of bone reaction in the surrounding parts of the vertebral body suggests a fatal event.

Additional possible evidence of trauma can be drawn from the high bilateral humeral asymmetry exhibited by the Gravettian skeleton Barma Grande 2 (Churchill and Formicola, 1997), from the bilateral absence of the lesser trochanter shown in the Late Epigravettian individual Arene Candide 2 (Formicola et al., 1990), and from the dislocation of the humerus affecting the right shoulder of Chancelade 1 (Dastugue and de Lumley, 1976). While in the first case a muscle lesion or innervation of the shoulder is a likely cause, the etiology responsible for the bilateral absence of the lesser trochanter is more problematic. A possible explanation is bilateral avulsion following trauma and consequent nonreattachment of the injured apophyses. Further, although difficult to prove, evidence of trauma is provided by the Gravettian adolescent from Arene Candide cave, "Il Principe." The hypothesis of trauma is based on lack of the left mandibular body, a bone generally subject to little postmortem deterioration, in a skeleton otherwise complete. In addition, there is bone loss in the left upper trunk close to the mandible (left scapula, left clavicle, and head of the left humerus). Moreover, Cardini (1942) suggested that the big lump of yellow ochre found below the right part of the mandible represents an expedient to hide the wound disfiguring the face of the adolescent (see Fig. 13).

Periodontopathies and alveolar changes, if present, affect only fully adult individuals showing marked dental wear (e.g., DV 16, Grotte des Enfants 4, Barma Grande 2) (Formicola, 1988; Formicola and Repetto, 1989; Trinkaus et al., 2005). Caries affect very few Gravettian remains (e.g., Cro-Magnon 4, Le Rois, Paglicci) (Borgognini et al., 1980; Brennan 1991; Trinkaus et al., 2005) and appear less sporadically in LUP remains in southwestern France (Brennan, 1991), and in the Italian Epigravettian (Villabruna 1, Romito, Ortucchio) (Parenti, 1960; Fabbri and Mallegni, 1988; Alciati et al., 2005). With few exceptions, oral health conditions remain good in the LUP. Roc del Migdia (Spain) (Turbón, 1989), showing multiple caries associated with antemortem tooth loss, probably represents an exceptional case.

The generally good health conditions observed in UP dentition also characterize other parts of the skeleton. Signs of infectious diseases are extremely infrequent. There is a case of otitis (DV 14) (Trinkaus et al., 2005) and of acute periostitis in Brno 2 (Oliva, 2000). Considering the severity of bony reaction and the fact that periostitis in Brno 2 affects the few preserved fragments of long bones (femur and ulna), this is likely a diffuse infectious process rather than localized infection. Localized periostitis, possibly of traumatic origin, has been observed in the Lagar Velho child radial shaft (Trinkaus et al., 2002) and in the tibial diaphysis of the LUP skeleton from Villabruna (Vercellotti et al., 2008). Brennan (1991) points out a general increase in frequency of periostitis in southwestern France during the LUP. Further evidence of infectious processes can be found in the alveolar region as a consequence of periapical lesions and abscesses following trauma or advanced wear (e.g., Grotte des Enfants 4) (Formicola and Repetto, 1989). Moreover, the well-known frontal and mandibular erosions exhibited by

Cro-Magnon 1 have been attributed by Dastugue (1967) to actinomycosis, a chronic disease caused by a bacterium (*Actynomices israelii*), originally affecting soft tissues and subsequently spreading to the underlying bone. Thillaud (1981–1982), however, disagrees with this diagnosis and refers the observed changes to a systemic disorder (histiocytosis X, a group of conditions characterized by proliferation of histiocytes). The scanty evidence of systemic or localized infective processes attests to the low pathogen load associated with a hunter–gatherer lifestyle (Roosevelt, 1984). Finally, while there is growing evidence of porotic hyperostosis at the end of the UP (Ascenzi and Balistreri, 1977; Brennan, 1991), and probably also among the Aurignacians from Mladeč (Teschler-Nicola et al., 2006) a likely, alternative explanation for infectious disease or malnutrition lies with hookworm or other parasite infestation.

The paleopathological literature also includes a possible case of hydrocephalia, affecting the child from Rochereil (Vallois, 1971, but see Tillier et al., 2001). Hydrocephalia is a disease of congenital or acquired origin whose earliest known example is found in the Qafzeh 12 child (Tillier et al., 2001). The Rochereil remains, which are attributed to the Magdalenian, show signs of postmortem trephination carried out from the internal part of the frontal bone in view of acquiring a rondelle. It is noteworthy that this practice is only known starting with the Neolithic.

Interestingly, the UP presents a number of cases of congenital or inherited diseases. The earliest evidence is provided by a congenital inner ear malformation affecting Mladeč 5, possibly resulting in hearing deficiency (Teschler-Nicola et al., 2006). Additional cases come from the triple burial of Dolní Vestonicě (Moravia) and the double child burial of Sunghir (Russia). The deformities affecting one of the skeletons from the triple burial, DV 15, have been attributed to chondrodysplasia calcificans punctata (Formicola et al., 2001). This diagnosis has been contested by Trinkaus et al. (2005), who attribute these changes to a congenital, unidentifiable, dysplasia. Main skeletal changes exhibited by DV 15 include asymmetric shortening and bowing of the femur, elongation of fibulae, and upper limb deformities, possibly of traumatic origin. A marked enamel hypoplasia on the first molars is probably associated with acute phase of the disease during early childhood. Despite these handicaps, DV 15 survived until adulthood, and upper and lower limb bone robusticity and articular changes attest to the active life of this individual (Trinkaus et al., 2001).

A further case of congenital disorder is shown by the 9- to 10-year-old girl found in the double burial from Sunghir (Sunghir 3) (Buzhilova, 2000; Kozlovskaya, 2000; Mednikova, 2000). As with Dolní Vestoniče, the burial is dated to the Gravettian. Both femora are deformed, short, and markedly anteroposteriorly bowed, while the remaining skeleton does not show any developmental disruption. The isolated nature of these deformities point to a nonspecific condition indicated in the orthopedic literature as congenital bowing of long bones, whose etiology may be found in disturbances of bone ossification, possibly in relation to the diabetic condition of the mother (Formicola and Buzhilova, 2004). If this is the case, Sunghir 3 would provide indirect evidence of diabetes in the Gravettian. Finally, Teschler-Nicola et al. (2006) briefly mention the marked curvature of the shaft in Mladeč 78's left femur described by Szombathy and strongly recalling that exhibited by Sunghir 3.

The Romito case (see Fig. 14) undoubtedly represents the most intriguing evidence of inherited disorders in the Paleolithic. Romito 2 is one of the Late Epigravettian skeletons found at the Romito site (Italy) (Bachechi and Martini, 2002). This young individual, buried with an adult female, was affected by an inherited form of acromesomelic dwarfism (Frayer et al., 1987, 1988). Romito 2 shows typical characteristics of dwarfism, including, among others, very short stature, extreme shortening of the distal upper limb, restricted elbow extension, cranial bossing, and constricted cranial base.

Another Late Epigravettian site (Arene Candide, Italy) provides further evidence of inherited disorders during the LUP. The diagnosed disorder in this case is an inherited form of rickets (X-linked familial hypophosphatemic rickets) (Formicola, 1995), characterized by long bone bowing deformities, abnormal tendon and ligament calcifications, and ossification disturbances. While the diagnosis concerns a male specimen (Arene Candide 3), this disorder could explain abnormalities characterizing additional, sometimes incomplete, remains from the same site.

These early examples of congenital disorders add interest to the history of the diseases and provide clues regarding the concepts shaping UP funerary behavior (Formicola, 2007) and on the survival of disabled individuals (Trinkaus et al., 2001). Due to their congenital nature, however, they are of minor importance for inferences on health conditions of past populations. To address this issue, we summarize the results provided by nonspecific stress indicators such as enamel hypoplasias and Harris lines. Enamel hypoplasias are infrequent and of low intensity in EUP material from southwestern France (Brennan, 1991) and Europe in general. This is also the case with the Moravian Gravettian remains (Trinkaus et al., 2005) and the Aurignacian sample from Mladeč (Teschler-Nicola et al., 2006). Only in two Gravettian cases does severity of enamel grooves point to a different pattern. Marked enamel defects are found in the first molars of DV 15 (Trinkaus et al., 2005) and in the canines of Grotte des Enfants 6 (see Fig. 15) (Alciati et al., 2005). While in the first case the defects are likely the effect of disturbances linked to the congenital disease affecting this specimen (see preceding text), there are no clues regarding etiology for the latter case. Brennan (1991) points out an increased frequency of hypoplasias between EUP and LUP. The intensity of hypoplasias, however, remains light in both periods. Moreover, while the age of appearance of the defects is generally around 4–5 years in Gravettian Moravia, it goes down to 2.5–3.5 years in the LUP in southwest France. In both cases, the authors suggest connections to nutritional stress associated with weaning. If this interpretation is correct, we should assume that the weaning age decreases over time or that it varies among geographic groups. Brennan's results suggest greater exposure to growth disruptions in the LUP than in the EUP. They also point out that severity of these stress indicators in the UP is much lower than that observed in more recent populations. Data reported on Italian UP remains (Alciati et al., 2005) confirm this finding. However, given the different protocols adopted for scoring stress indicators, for the sake of homogeneity, in Table 2, we report only the results observed by Brennan (1991).

In conclusion, specific and nonspecific stress indicators emphasize the concept, evidenced by previous studies, of UP groups as healthy people, but also point to differ-

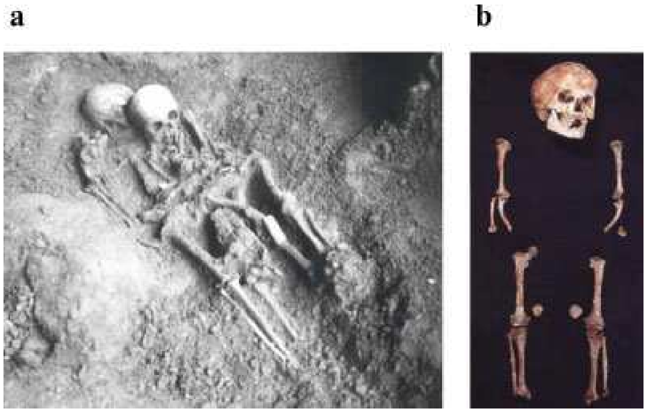

Fig. 14. (a) Double burial from Romito (Italy) at the moment of discovery. The dwarf is on the right. (b) Romito 2, the adolescent, is affected by acromesomelic dwarfism. (Romito burial: From Bachechi and Martini, 2002; courtesy of F. Martini. Romito 2: From Frayer DW, et al., 1988. A case of chondrodystrophic dwarfism in the Italian Late Upper Paleolithic. *American Journal of Physical Anthropology* 75:549–565. © 1988 Wiley-Liss.). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

ences through time. In particular, LUP samples show an increase in frequency but not in severity of enamel hypoplasia and Harris lines, compared to EUP (Brennan 1991 and personal observations). Caries become less sporadic in the LUP, indicating a possible increase in consumption of vegetal food rich in carbohydrates. The most intriguing aspect of trauma comes from the clear evidence of wounds resulting from projectiles. Although these cases could be incidental, the hypothesis of increased conflict among LUP groups should be taken into account. This is certainly the case for the large LUP sample from Jebel Sahaba (Sudan), in which numerous skeletons exhibit signs of violence in the form of fractures and cut marks (Anderson, 1968). Beyond these considerations that, as with congenital diseases, have mostly a sociological and behavioral value, the data presented here indicate a slight decline in health status during the LUP, a pattern supported by the decrease in stature during this time period (see preceding text).

Diet

Traditionally, diet reconstruction in past populations has been based on analysis of faunal and vegetal remains as well as remains of material culture. These analyses have been supplemented starting from the middle of the past century by information drawn from dental

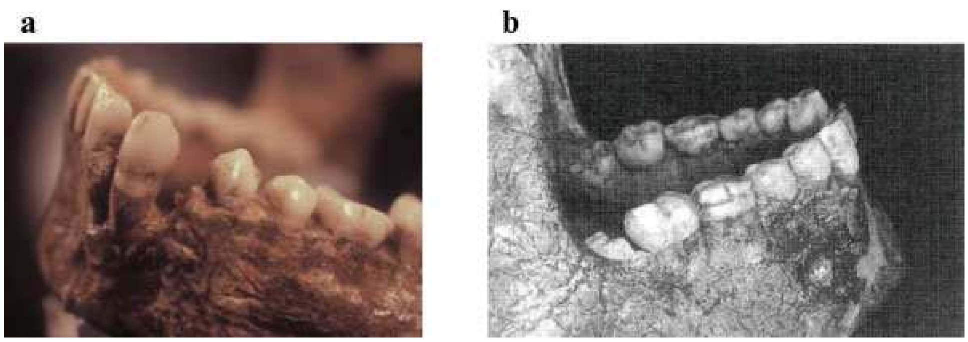

Fig. 15. Dental enamel hypoplasia of (a) Grotte des Enfants 6, and (b) Dolní Vestonice 15. These are two rare Gravettian examples of marked hypoplasia. (Courtesy of the Musée d'Anthropologie Préhistorique de Monaco and Institute of Archaeology AS Dolní Vestonice, respectively). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

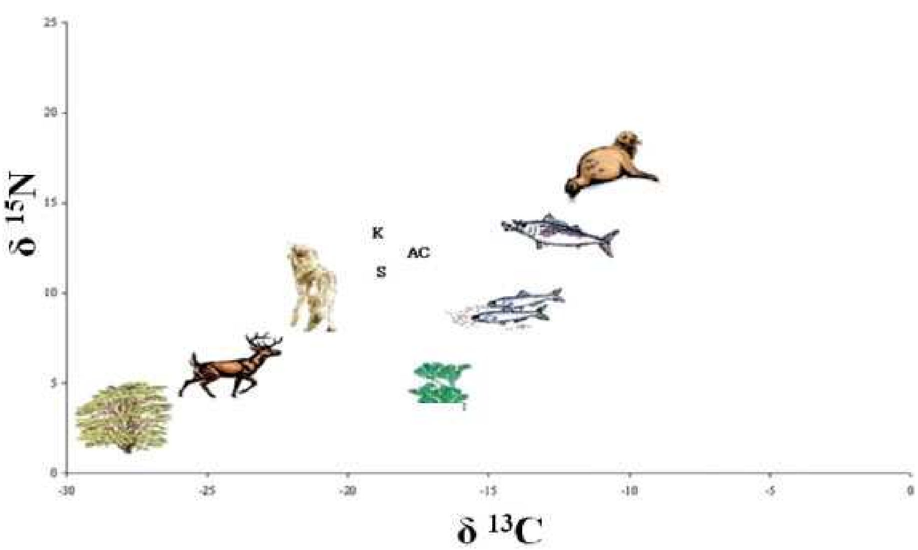

Fig. 16. Carbon (δ^{13}C) and nitrogen (δ^{15}N) stable isotope values for three Gravettian specimens: Sunghir 1 (S), Kostenky 4 (K), and Arene Candide 1 (AC). The values of the two isotopes place these individuals high in the food web and suggest significant consumption of herbivores (S) and of freshwater resources (K) as well as a contribution from marine resources to the diet of AC. (Adapted from Goude, 2007; courtesy of G. Goude). [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

TABLE 2. Frequency of stress indicators in EUP and LUP from southwest France, (from Brennan, 1991)

	EUP		LUP	
Stress Indicators	*N*	%	*N*	%
Hypoplasias	35	16.2	68	29.4
Harris lines	12	33.3	25	48.0
Periostitis	44	2.3	102	11.8
Arthritis (adults)	9	11.1	25	16.0
Parietal porosity	20	30.0	39	25.0
Cribra orbitalia	4	0.0	19	10.5
Caries	35	2.9	68	7.3
Periodontal disease (adults)	7	42.9	25	12.0

wear and microwear. Orientation, typology, and dimension of the microconfigurations provide information on the biomechanics of mastication and on the range and abrasive content of the material exploited. Lalueza et al. (1996), for instance, document wear patterns in the EUP and LUP that are indicative of a mixed diet, more abrasive and probably richer in vegetal components than that of Neandertals. These differences, however, are poor indicators of the nature of abrasive components of the diet and correlate to some extent with ecogeographic and climatic conditions (Peréz-Peréz et al., 2003; El Zaatari, 2007). In addition, microwear texture left on teeth may result both from alimentary and nonalimentary activities (King et al., 1999) and provide information on the last masticatory events, not on the diet in its broad sense.

Beginning in the 1980s, analyses of chemical composition of bones have increasingly supplemented other sources of information on dietary reconstruction. Bone is made of inorganic (hydroxyapatite) and organic (collagen) components whose chemical composition varies according to the kind of food ingested. Francalacci (1989) analyzed trace elements in LUP and Neolithic samples from western Liguria (Italy) for inferences on changes in diet composition at the transition from a hunting and gathering economy to one of food production. Results suggest that the meat/vegetal ratio did not change, probably due to the replacement of wild with domesticated fauna and vegetals. High intraindividual variation and diagenetic contamination, however, suggest that the results from trace element analyses should be cautiously regarded and checked against other sources of information (Francalacci, 1989, 1990).

More recently, chemical analyses have shifted from inorganic to organic components of the bone by measuring the ratios of carbon ($\delta\ ^{13}$C) and nitrogen ($\delta\ ^{15}$N) stable isotopes. This method allows investigation of the sources of proteins during the last years of life and is particularly suitable in distinguishing marine from terrestrial sources and in positioning an organism within the food web (see Fig. 16). Comparative analyses of carbon and nitrogen stable isotope values (Richards et al., 2001) demonstrate increased diet breadth after the Middle Paleolithic. In addition to the large herbivores that formed a primary source of food for Neandertals, the EUP sample shows evidence of significant consumption of freshwater resources, potentially including aquatic birds, mammalian predators, mollusks, and fish. According to Richards and coworkers, these sources form an important component of the diet, ranging from about 50% at Kostenki to 25–50% in other continental sites, such as Mal'ta, Dolní Vestoniče, and Brno. Zoo-archeological evidence from Gravettian sites from southwestern Germany (Brillenhohe, Hohefels, Geissenklosterle) (Hahn, 2000), also points to the importance of these resources. Among Gravettian samples, isotope evidence of consumption of terrestrial herbivores is particularly strong at Sunghir and Paviland. Conforming to its coastal location, the latter specimen also shows that marine resources, although of minor importance (around 10–15%) were part of the diet, while data from the nearby site of Eel Point are unclear but seem to exclude marine food intake (Schulting et al., 2005). Consumption of marine food at coastal sites from Liguria is documented by different sources of data. Isotope analyses carried out on the Gravettian specimen "Il Principe" from Arene Candide indicate that about 20%–25% of his diet included marine food (Pettitt et al., 2003), while consumption of the same kind of resources is inferred from zoo-archeological data in the Aurignacian of Riparo Mochi (Kuhn and Stiner, 1998). Although Drucker and Bocherens (2004) question the conclusions of Richard et al. (2001), based on the lack of isotopic data on contemporary fauna of known diet, zoo-archeological information concurs in suggesting the increased importance of aquatic resources as well as of terrestrial small game during the EUP (Stiner, 2001; Kuhn and Stiner, 2006).

Marine resource exploitation seems to have increased during the LUP, as shown by isotope data from coastal sites in North Wales (UK) (Richards et al., 2005), where the aquatic component, which comprised approximately 30% of protein intake, may have resulted from hunting marine mammals such as seals. Richards and coworkers stress, however, that isotope data do not allow distinction of seasonality in consumption of these resources. It is therefore not possible to tell whether these populations were mobile or settled yearlong along the coast. Bocherens and Drucker (2006) question these inferences on mobility patterns, suggesting that anadromous fish, such as salmon and freshwater fish, could represent an alternative explanation to consumption of seals.

The importance of herbivores in the LUP diet emerges from noncoastal sites in southwest England (Gough's Cave and Sun Hole Cave) (Richards et al., 2000) as well as in Magdalenian and Solutrean French remains (Hayden et al., 1987). The scarcity of marine resources in middle Magdalenian southwestern France is stressed by zoo-archeological findings (Le Gall, 1992) and by stable isotope results obtained from Abri Faustin (Drucker et al., 2005) and Saint Germain-la-Rivière (Drucker and Henry-Gambier, 2005).

In conclusion, the expansion of alimentary strategies that apparently characterized the EUP continued in the LUP. The bulk of the data also suggests increased reliance on marine foods that would become the main alimentary source for Mesolithic populations from coastal Europe (Richards and Hedges, 1999; Schulting and Richards, 2001).

Physical activity

As with the skull, the postcranial skeleton of UP modern humans is generally said to be less robust than that of Neandertals, particularly with respect to the upper limb. This process of gracilization in Late Pleistocene Europe, including decreased robusticity and stature, has been linked to improvement in hunting technology and concomitant decline in prey size (Brose and Wolpoff, 1971; Frayer, 1981). According to this model, better hunting technology that allowed more distance between prey and hunter as well as the reduction in the size of prey, led to a decrease in risk and strength requirements associated with hunting (Frayer, 1980, 1981, 1984). Resulting relaxation of selection for large body size is presumed to have been the catalyst in the process. Furthermore, a larger decrease in male rather than female strength resulted in reduced sexual dimorphism in body size. The rationale behind this decrease in sexual differences is based on hunter–gatherer ethnographic evidence of higher male involvement in hunting activities (Frayer, 1980; Ruff, 1987; Kelly, 1992). Churchill's (1994, 1996) analysis of late Pleistocene hominin body form suggests that, while there was some correlation and integration between certain skeletal elements, the evolution of body size was characterized by a substantial amount of plasticity. For instance, although stature decreased between the EUP and LUP, evidence of increased upper limb muscularity and robusticity during that time period (Jacobs, 1985; Churchill, 1994) suggests that models linking improved efficiency of hunting technology with reduced body size and strength may be oversimplified (Bridges, 1995).

Factors affecting body size and robusticity[2] include nutritional, mechanical, climatic, genetic, and hormonal effects. Pearson (2000), for example, argued that the increased postcranial robusticity he observed between EUP and LUP groups was linked to cold adaptation in the latter. Given archeological evidence of decreased mobility and improved subsistence technology during the LUP, and the remarkable ability of bone to respond to levels of mechanical loads, it is reasonable to assume that these changes affected the postcranial skeleton in some way. Numerous studies have demonstrated a clear correlation between physical activity and skeletal robusticity (see review in Ruff et al., 2006). This association has proven quite fruitful in tracking patterns of behavior in past populations, in particular when associated with marked subsistence shifts (involving marked changes in levels of physical activity), such as the shift from foraging to agriculture (Ruff and Hayes, 1983; Ruff and Larsen, 1990). Biomechanical analyses of long bone diaphyses show that localized strains engendered by mechanical loads alter the amount and distribution of cortical tissue within the cross-section. This relationship between bone adaptive remodeling and behavior has allowed for testing of hypotheses regarding specific subsistence activity involving the lower limb (Holt, 1999, 2003; Stock and Pfeiffer, 2001) and upper limb use (Churchill 1994; Churchill et al., 2000; Weiss, 2003). While unresolved issues remain regarding the exact process by which adaptive remodeling occurs, as well as the effect of factors such as body mass and proportions and age (see reviews in Pearson and Lieberman, 2004; Ruff et al., 2006), numerous studies have confirmed the validity of the structural approach to the study of robusticity. A recent study (Stock and Shaw, 2007) confirms that a cross-sectional geometric approach that includes modeling of periosteal contours produces more accurate estimates of diaphyseal mechanical strength than do external shaft diameter and circumference.

Holt (1999, 2003) tested archeological evidence that environmental impacts of the LGM resulted in decreased mobility during the LUP. Previous studies (Brock and Ruff, 1988; Ruff and Larsen, 1990) showed that reduced

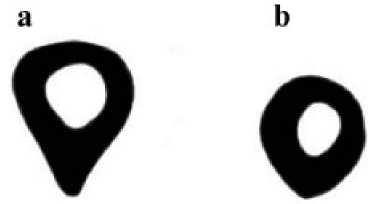

Fig. 17. Tracings of two midshaft femoral cross-sections. (a) EUP (Parabita 1). (b) LUP (Arene Candide 10). Note the difference in shape reflecting higher antero–posterior bending rigidity of EUP femur.

mobility associated with the shift from foraging to agricultural economies resulted in a decreased cortical thickness as well as an increased cross-sectional circularity of the midshaft femur, as an effect of the decrease in anteroposterior (AP) mechanical loads. In a recent study of the effects of hunter–gatherer terrestrial and marine mobility on lower limb robusticity, Stock (2006) confirmed strong correlations between midshaft femur cross-sectional shape and terrestrial mobility. Holt's analysis showed a significant decrease in femoral diaphyseal robusticity between the EUP and LUP, particularly in midshaft femur AP bending rigidity, in keeping with expectations of reduced locomotor behavior (see Fig. 17). While midshaft femur circularity increases further after the end of the Last Glacial, during the Mesolithic, these results show that significant changes in mobility levels occurred during the UP. It should be noted that the effect of topography on lower limb robusticity was not included in the study. Evaluation of the respective effects of subsistence and topographic relief on lower limb cross-sectional properties in Ameridian prehistoric populations shows that topography had a more significant effect on these properties than did subsistence (Ruff, 2000a). While these results suggest that both subsistence-related activity and terrain should be taken into account in biomechanical analyses of the lower limb, the fact that samples in Holt's study come from all areas of Europe increases confidence that terrain did not have a confounding effect.

Femoral anterior curvature also relates to physical activity. Shackelford and Trinkaus (2002) showed that recent urban populations exhibit significantly lower anterior femur curvature than less industrialized groups, a difference they attributed to degrees of sedentism and mechanization. Furthermore, they found that a Middle Paleolithic sample, comprised of samples from Skhul and Qafzeh and EUP Europeans, exhibited higher anterior curvature than an LUP sample, in agreement with diaphyseal robusticity results discussed in preceding text.

Finally, Trinkaus's (1993) analysis of femoral neck-shaft angle (NSA) in Late Pleistocene hominins showed that EUP femora have a lower NSA than LUP femora. In recent populations, Trinkaus found that the NSA increases from foragers to agriculturalists to urbanized groups, in tandem with general trends in robusticity. Since the NSA reflects levels of locomotor activity during ontogeny (Houston and Zaleski, 1967; Trinkaus, 1993), a low angle in EUP femora suggests high mobility during development. Interestingly, two of the femora included in Holt's (2003) EUP sample belonged to relatively young males, DV 14 and AC 1 (Il Principe). While the two individuals had large medullary cavities typical for their

[2]Here, robusticity refers to mechanical strength of postcranial skeletal elements relative to body size.

TABLE 3. Humeral diaphyseal cross-sectional robusticity measures[a]

	EUP		LUP	
Males	Right (N = 9)	Left (N = 11)	Right (N = 13)	Left (N = 10)
CA[b]	222.9[c] (34.9)	174.5 (28.4)	275.6 (40.5)	227.9 (52.0)
J[d]	1466.7 (322.3)	921.8 (294.2)	1811.8 (1309.9)	1300.0 (490.3)
Ix/Iy[e]	130.8 (16.5)	126.4 (16.5)	118.6 (15.6)	121.1 (25.0)
Females	Right (N = 2)	Left (N = 4)	Right (N = 8)	Left (N = 7)
CA[b]	213.2 (19.1)	183.2 (28.3)	214.2 (43.8)	196.3 (29.7)
J[d]	1128.1 (95.1)	953.2 (213.3)	1171.9 (388.1)	1178.6 (196.2)
Ix/Iy[e]	120.3 (3.0)	124.9 (18.7)	128.0 (12.7)	135.8 (12.0)

[a] Data from Churchill et al. (2000) and personal communication.
[b] CA, Cortical area, standardized to humeral length^2 ($\times 10^5$).
[c] Mean, with standard error in parentheses.
[d] J, Polar moment of area standardized to humeral length^4 ($\times 10^9$).
[e] Ix/Iy, Ratio of antero-posterior to medio-lateral bending moments.

age, the shape of the cross-section shows the AP elongation characteristic of EUP adult femora, also suggestive of high levels of adolescent mobility.

A different pattern emerged from analyses of upper limb robusticity (Churchill, 1994; see also Churchill et al., 2000). In contrast with the lower limb data, the LUP specimens showed a significant increase in humeral strength over their EUP counterparts (Table 3). For both females and males, LUP humeri showed stronger resistance to bending and torsional stresses, although small sample sizes for females prevented statistical tests. It should be noted that humeral length was used to scale the upper limb cross-sectional dimensions to control for body size, and thus that increased diaphyseal strength may be an artifact of the shorter limb of LUP groups. It would be interesting to see whether these results hold true when robusticity measures are scaled to body mass estimates using either femoral head or a combination of stature and biiliac breadth (Ruff, 2000b).

Asymmetry in UP humeral robusticity is quite high, even relative to recent marine foragers such as the Aleut, who otherwise have strong humeri, reflecting their extensive use of boats and kayaks over open waters (see Fig. 18). The high asymmetry measures in UP males, particularly in the EUP, probably reflect the repeated unimanual use of hunting weapons (Churchill et al., 2000; Schmitt et al., 2003). One further change relates to cross-sectional shape. EUP humeri exhibit a much more AP-elongated shape, while LUP humeri are more circular (Table 3). This difference in shape may reflect a shift in hunting technology involving handheld thrusting spears that engendered AP bending stresses, to increased reliance on thrown projectiles that would have produced more evenly distributed stresses (Churchill et al., 2000; Schmitt et al., 2003).

Analysis of cross sectional properties of limb bones provides additional interesting information concerning sexual dimorphism. While males show higher levels of lower limb robusticity than females, in keeping with evidence from recent foragers (Ruff, 1987) and early agriculturalists (Marchi et al., 2006), one surprising result from this analysis is the lack of a clear trend in sexual dimorphism in either bone strength or shape (Holt, 2003). Decreasing mobility in recent populations has generally been associated with reduced sexual dimorphism in lower limb robusticity, reflecting convergence between male and female locomotor activity (Ruff, 1987; Benfer, 1990). While small female sample sizes, particularly for the EUP, may contribute to the lack of a trend in sexual dimorphism, the degree of sexual dimorphism in lower limb robusticity in the LUP may also reflect increased involvement of females in subsistence activity, in the context of resource intensification and diversification in this time period (Mellars, 1985; Straus, 1993; Stiner et al., 1999). This interpretation finds some support in the upper limb data (Churchill et al., 2000). LUP female humeri show increased torsional bending strength over EUP females, but small sample sizes prevent meaningful statistical tests. Interestingly, however, both EUP and LUP females exhibit much higher upper limb robusticity asymmetry than recent foragers. These patterns of upper and lower bone strength suggest that sexual division of labor, universal among recent foragers (Kuhn and Stiner, 2006), may not have been part of UP organizational strategies. An alternative explanation is

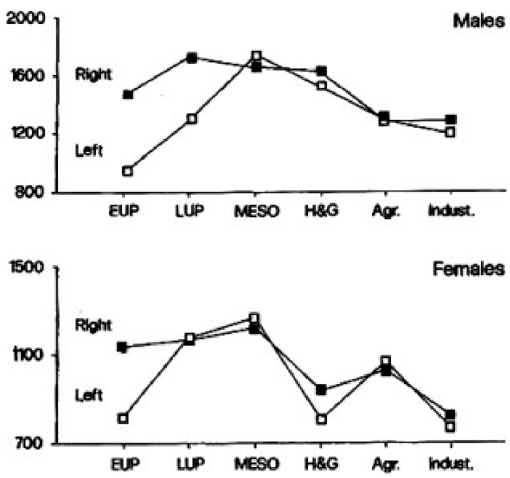

Fig. 18. Temporal change in size-standardized bone strength (J, or polar moment of area) for right and left humerus, by sex, for EUP, LUP, Mesolithic, and two recent hunter–gatherers (Aleut and Jomon), agriculturalists, and industrialized groups. (Data and figure from Steve Churchill, in Holt et al., 2000).

that males and females engaged in different activities that resulted in similar loading patterns (Collier, 1993).

Thus, the limb robusticity evidence reveals a more complicated picture than that painted by general body size. The rapid development in the LUP of increasingly sophisticated and diverse foraging technology did not result in upper limb gracilization. When placed in the ecological context of environmental degradation and decreased resource availability that followed the LGM, it appears that the accelerated development in subsistence technologies and increased upper limb robusticity may have been correlated responses to resource stress (Churchill et al., 2000). Modern ethnographic data shows that increased resource patchiness generally results in higher mobility (Kelly, 1995), particularly at higher latitudes. In the case of the LUP, however, loss of large areas to glacial advances produced a pattern of population packing that limited territorial expansion and mobility as adaptive responses.

Limb robusticity data from other regions of the world suggest that some of the changes that took place between EUP and LUP in Europe also characterized these areas. Studies of upper limb robusticity, for instance, also show marked bilateral asymmetry in some Levantine males, such as Ohalo II and Wadi Mataha (F-81) (Nadel and Hershkovitz, 1991; Stock et al., 2005), pointing, as in Europe, to the use of unimanual hunting technology. This conclusion is reinforced by evidence, in Levantine Epipaleolithic groups, of pronounced attachment scars for muscles involved in throwing activities (Peterson, 1997). Shackelford's (2007) comparisons of diaphyseal robusticity in LUP samples from the Nile valley, Mediterranean, and Asia show that, relative to the European EUP, all LUP groups exhibit lower femoral bending rigidity and higher femoral and humeral circularity. Patterns of asymmetry diverge somewhat from the European pattern, in that robusticity of the right humerus decreases in all LUP samples, while strength on the left side increases, resulting in lower levels of asymmetry. It is unfortunate that the paucity of well-preserved EUP skeletons outside of Europe prevents regional temporal comparisons. However, Shackelford's study suggests that the increase in population density, reduced mobility and increased diet breadth that occurred world-wide as a result of LGM-related declining temperatures and aridity (Bar-Yosef, 1996; Close and Wendorf, 1990; Hershkowitz et al., 1995; Kershaw et al., 2001) affected human subsistence behavior in similar ways.

CONCLUSIONS AND FUTURE DIRECTIONS

In stressing the divisive effects of the LGM on UP systems, we have undoubtedly been guilty somewhat of "throwing large blankets over the past" (Roebroeks and Corbey, 2000, p. 82). While the climatic impact and consequent environmental upheavals are quite real, biological and behavioral responses undoubtedly differed among groups. Neither archeological nor skeletal records do justice to this variability, and only permit us to address general patterns of biocultural adaptations, in as much as they reflect behaviors that are repetitive and cumulative (Gamble, 1986).

Articulation of climatic, ecological, archeological, and skeletal records shows that the culmination around 20 kya of progressive climatic degradation initiated 60 kya before had a profound impact on the lives of UP hunter–gatherers. This impact was felt on populations worldwide, although its magnitude was more marked in Europe. While technological innovations in response to environmental stress occured occasionally as early as 70 kya (Deacon and Deacon, 1999; Yellen et al., 1995), the remarkable acceleration after 20 kya in cultural development in Europe attests to increasing reliance on and mastering of technology as a way to "push back" against environmental challenges. This behavioral change might be reflected in the general decline in craniodental robusticity (Frayer, 1978). However, given the lack of correspondence between mechanical strain and facial structures such as the supraorbital torus (Hylander et al., 1991; Antón, 1994), as well as evidence that cranial shape and size may be affected significantly by development, the role of morphological integration in the ontogeny of cranial robusticity must be addressed. This is particularly important since many of the traits used to reconstruct phylogenetic relationships relate to cranial robusticity in some way.

Postcranial robusticity also reflects intensification of technology and change in organizational strategies. Overall levels of foraging mobility declined as a consequence of territorial contraction, resulting in a reduction in lower limb robusticity in LUP groups worldwide (Holt, 2003; Shackelford, 2007). Upper limb data provide a more complex picture. In more temperate regions of Asia and Africa, upper limb robusticity generally declined relative to EUP European groups, paralleling craniodental changes (Shackelford, 2007). In Europe, however, LUP males exhibited stronger humeri than their predecessors (Churchill et al., 2000). This picture of decreasing lower limb and increasing upper limb strength underscores the importance of particularistic models in studies of morphology (Churchill, 1998).

Decoupling of general body size variables, such as stature, from measurements that reflect localized mechanical loads has permitted researchers to address patterns of specific behaviors (Trinkaus et al., 1994; Ruff, 2000a; Churchill et al., 2000; Holt, 2003). For instance, while LUP males showed similar or larger levels of humeral strength relative to EUP, the increased cross-sectional circularity of their humeri and continued high asymmetry suggest a shift from thrusting to throwing hunting technology, a change observed in many LUP groups and supported by the archeological record (Churchill et al., 2000; Schmitt et al., 2003; Stock et al., 2005; Shackelford, 2007). Application of mechanical models will be important in understanding the effects of Tardiglacial conditions on human activity patterns.

Results from haplotype lineage analysis in modern populations (Richards et al., 2000; Torroni et al., 2001) have provided exciting new prospects regarding both the relationship between UP and extant populations as well as population movements during the UP itself. Studies on the Y chromosome and mtDNA point to two separate founding events, possibly linked to the diffusion of the Aurignacian and of the Gravettian in Europe (Torroni et al., 1998; Semino et al., 2000). An important, but challenging, problem will be to correlate the origins of these genetic lineages with archeological evidence of population movements. Evidence from ancient DNA (e.g., Caramelli et al., 2003) will also be important. For both ancient and modern DNA approaches, main obstacles remain the limitations of using haploid-linked markers and the lack of understanding about the effects of demographic dynamics (contractions, expansions, backward migrations) on genetic variability, and about the reliability of the assumption that mutations accumulate at a constant rate.

A major evidence of continuous biological adaptation in the UP is provided by the appearance of cold-adapted body proportions in LUP European groups (Holliday, 1997). The challenge will be to explain why limb proportions did not change accordingly. The notable decrease in stature in these populations (Formicola and Giannecchini, 1999) is another difficult change to explain. Given that tall stature in the EUP, in particular long legs, may have reflected tropical origins, high-quality diet, adaptation to high mobility, and/or elevated levels of gene flow, it is not surprising that EUP populations had long legs. While reasons for the marked reduction of stature are not fully understood, an adaptive process involving reduced body size and hence metabolic and nutritional demands may have played an important role (Formicola and Holt, 2007).

Reduction in stature and paleopathological data concur in indicating a slight decline in quality of life during the LUP. While the scarcity of systematic studies suggests caution about general conclusions, our survey shows that, relative to Holocene populations, particularly those relying on food producing economies, UP groups enjoyed good health and nutritional conditions. Surprisingly, however, both EUP and LUP samples exhibit a number of individuals affected by congenital and rare disorders. This finding is unlikely to reflect chance preservation of anomalous individuals. Instead, anomalous individuals, spectacular burial patterns, burial composition, and hints about genetic relationships among members of multiple burials (Formicola, 1988; Alt et al., 1997) point to selective funerary practices and provide clues about concepts of life and death among these populations (Pettitt, 2006; Formicola, 2007).

We began our paper by quoting Plato, who reminisces about the distant past when humans were content to live off what nature offered them. While it is true that UP subsistence was based on foraging and adaptive strategies that were often responses to environmental pressure, it is also abundantly clear that UP populations differed from previous groups in their ability to dominate their environment. Gravettian organizational systems exhibit hallmarks of modernity, such as extensive reliance on the cushioning effects of complex social networks. Recurrence over thousands of kilometers of stylistic elements repeated in objects and funerary practices attests to the unprecedented scale of these networks. The composition of these burials and the richness of associated grave goods reveal the extraordinary complexity of EUP social structure, symbolism, and beliefs. When groups were finally forced to retreat by increasingly harsh conditions, this finely tuned system ceased to exist, bringing an end to what has been called the "Golden Age" of hunters (Mussi et al., 2000). Articulation of skeletal data and archeological context allows us to track some of the ways in which UP people adapted, biologically and by choice.

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APPENDIX

TABLE A1. The EUP and LUP skeletal sample by region, country, and site

Samples[a]	Country	Age/Sex	Remains	Date[b]	References
		Early Upper Paleolithic			
Europe					
Krems Wachtberg 1	Austria	Newborn	Complete skeleton	26,580 ± 160	Einwögerer et al., 2006
Krems Wachtberg 2	Austria	Newborn	Complete skeleton	26,580 ± 160	Einwögerer et al., 2006
Krems Wachtberg 3	Austria	Newborn	Complete skeleton	26,580 ± 160	Einwögerer et al., 2006
Brno 2	Czech Republic	M	Incomplete skeleton	**23,680** ± **200**	Jelínek, 1959, Pettitt and Trinkaus, 2000
Dolni Věstonice 3	Czech Republic	F	Complete skeleton	25,950 + 630/− 580	Trinkaus and Jelínek, 1997; Trinkaus and Svoboda, 2006
Dolni Věstonice 13	Czech Republic	M	Complete skeleton	26,640 ± 110	Svoboda, 1995; Trinkaus and Svoboda, 2006
Dolni Věstonice 14	Czech Republic	Adolescent M	Complete skeleton	26,640 ± 110	Svoboda, 1995; Trinkaus and Svoboda, 2006
Dolni Věstonice 15	Czech Republic	?	Complete skeleton	26,640 ± 110	Svoboda, 1995; Trinkaus and Svoboda, 2006
Dolni Věstonice 16	Czech Republic	M	Complete skeleton	25,570 ± 280	Svoboda, 1995; Trinkaus and Svoboda, 2006
Pavlov 1	Czech Republic	M	Incomplete skeleton	26,000–25, 000	Svoboda, 1997
Mladeč 1	Czech Republic	F	Cranium	**31,190** + **400** − **390**	Wild et al., 2005; Teschler-Nicola, 2006
Mladeč 2	Czech Republic	F	Calvaria	**31,320** + **410** − **390**	Wild et al., 2005; Teschler-Nicola, 2006

(*continued*)

TABLE A1. (Continued)

Samples[a]	Country	Age/Sex	Remains	Date[b]	References
Mladeč 5	Czech Republic	M	Calvaria		Teschler-Nicola, 2006
Abri Pataud 1	France	F?	Skull		Billy, 1975
Cro-Magnon 1	France	M	Skull	27,680 ± 270	Vallois and Billy, 1965; Henry-Gambier, 2002; Gambier et al., 2006
Cro-Magnon 2	France	F?	Cranium	27,680 ± 270	Vallois and Billy, 1965; Henry-Gambier, 2002
Cro-Magnon 3	France	M	Cranium	27,680 ± 270	Vallois and Billy, 1965; Henry-Gambier, 2002
Paviland 1	England	M	Postcranial bones	**26,350 ± 550**	Aldhouse Green, 2000
Arene Candide 1	Italy	Adolescent M	Complete skeleton	**23,440 ± 190**	Sergi et al., 1974; Pettit et al., 2003
Barma Grande 1	Italy	M	Incomplete skeleton		Graziosi, 1942
Barma Grande 2	Italy	M	Complete skeleton		Formicola, 1988
Barma Grande 3	Italy	Adolescent F?	Incomplete skeleton		Formicola, 1988
Barma Grande 4	Italy	Adolescent F?	Incomplete skeleton		Formicola, 1988
Barma Grande 5	Italy	M	Incomplete skeleton		Verneau, 1906
Barma Grande 6	Italy	M	Postcranial bones	**24,800 ± 800**	Massari, 1958; Formicola et al., 2004
Caviglione 1	Italy	F	Complete skeleton	28,780 ± 560 20,220 ± 260	Rivière, 1887; Henry-Gambier, 2001
Bausu da Ture 2	Italy	M	Incomplete skeleton		Verneau, 1906
Grotte des Enfants 4	Italy	M	Complete skeleton		Verneau, 1906
Grotte des Enfants 5	Italy	F	Complete skeleton		Verneau, 1906
Grotte des Enfants 6	Italy	Adolescent M?	Complete skeleton		Verneau, 1906
Grotta Paglicci 12	Italy	Adolescent M	Complete skeleton	24,720 ± 420	Mallegni and Parenti, 1972; Palma di Cesnola, 2002.
Grotta Paglicci 25	Italy	F	Complete skeleton	23,470 ± 370; 23,040 ± 380	Mallegni et al., 1999; Palma di Cesnola, 2002
Grotta delle Veneri (Parabita) 1	Italy	M	Postcranial bones		Mallegni et al., 2000
Grotta delle Veneri (Parabita) 2	Italy	F	Postcranial bones		Mallegni et al., 2000
Ostuni 1	Italy	F	Complete skeleton	24,410 ± 320	Vacca and Coppola, 1993
Ostuni 1b	Italy	Fetus	Complete skeleton	24,410 ± 320	Vacca and Coppola, 1993
Ostuni 2	Italy	Adult	Incomplete skeleton		Alciati et al., 2005
Lagar Velho 1	Portugal	Child	Complete skeleton	25,000–24,000	Zilhão and Trinkaus, 2002
Cioclovina 1	Romania	F?	Calvaria	**28,510 ± 170**	Soficaru et al., 2007
Muierii 1	Romania	?	Incomplete skeleton	**29,930 ± 170; 30,150 ± 800**	Soficaru et al., 2006
Oase 2	Romania	?	Cranium	34,950 + 990/ − 890	Trinkaus et al. 2003
Sunghir 1	Russia	M	Complete skeleton	**22,930 ± 200**	Pettit and Bader, 2000; Alexeeva and Bader, 2000
Sunghir 2	Russia	Adolscent M	Complete skeleton	**23,830 ± 220**	Pettit and Bader, 2000; Alexeeva and Bader, 2000
Sunghir 3	Russia	Child F	Complete skeleton	**24,100 ± 240**	Pettit and Bader, 2000; Alexeeva and Bader, 2000
Sunghir 5	Russia	F?	Cranium		Alexeeva and Bader, 2000
Kostenky 3	Russia	Child	Complete skeleton	>25,700	Hoffecker, 2002
Kostenky 4	Russia	Child	Complete skeleton	**21,020 ± 180; 19,830 ± 120**	Sinitsyn, 2004
Africa					
Nazlet Khater 1	Egypt	M	Incomplete skeleton	37,570 + 350 − 310	Thoma, 1984; Vermeersch, 2002
Asia (Western)					
Nahal Ein Gev	Israel	F	Complete skeleton	22,000–25,000	Arensburg 1977; Hershkovitz et al., 1995
Asia (Eastern) and Australia					
Kow Swamp	Australia	M	Seven individuals (crania and postcranial bones)	15,000–9,000 22,000–19,000	Thorne 1969, 1971; Stone and Cupper, 2003

(*continued*)

TABLE A1. (Continued)

Samples[a]	Country	Age/Sex	Remains	Date[b]	References
Kow Swamp	Australia	F	Incomplete skeleton	15,000–9,000 22,000–19,000	Thorne 1969, 1971; Stone and Cupper, 2003
Kow Swamp	Australia	Juveniles	Six individuals (crania and postcranial bones)	15,000–9,000 22,000–19,000	Thorne 1969, 1971; Stone and Cupper, 2003
Kow Swamp	Australia	Unsexed	Two individuals (crania and postcranial bones)	15,000–9,000 22,000–19,000	Thorne 1969, 1971; Stone and Cupper, 2003
Lake Mungo 1	Australia	F	Incomplete skeleton (cremated)	40,000 ± 2,000	Bowler et al., 1970, 2003; Thorne 1971
Lake Mungo 3	Australia	M	Incomplete skeleton	**62,000 ± 6,000** 40,000 ± 2,000	Thorne et al., 1999; Bowler et al., 2003
Luijiang	China	M	Incomplete skeleton	30,000–60,000	Wu, 1982
Tianyuan Cave 1	China	?	Incomplete skeleton	**34,430 ± 510**	Shang et al., 2007
Zoukoudian Upper Cave (101)	China	M	Cranium	29,000–34,000	Wu and Poirier, 1995
Zoukoudian Upper Cave (102)	China	F	Cranium	29,000–34,000	Wu and Poirier, 1995
Zoukoudian Upper Cave (103)	China	F	Cranium	29,000–34,000	Wu and Poirier, 1995
Minatogawa 1	Japan	M	Incomplete skeleton	18,250 ± 650–16,600 ± 300	Suzuki and Hanihara, 1982
Minatogawa 2	Japan	F	Incomplete skeleton	18,250 ± 650–16,600 ± 300	Suzuki and Hanihara, 1982
Minatogawa 3	Japan	F	Incomplete skeleton	18,250 ± 650–16,600 ± 300	Suzuki and Hanihara, 1982
Minatogawa 4	Japan	F	Incomplete skeleton	18,250 ± 650–16,600 ± 300	Suzuki and Hanihara, 1982
Niah Cave	Malaysia	F?	Cranium	**37,000–35,000**	Barker et al., 2007
Fa Hien 2	Sri Lanka	Child	Incomplete skeleton	30,060 ± 410	Kennedy and Zahorsky, 1997
Moh Khiew 1	Thailand	F	Incomplete skeleton	25,800 ± 600	Pookajorn, 1996; Matsumura and Pookajorn, 2005
			Late Upper Paleolithic		
Europe					
Bruniquel 24	France	F	Complete skeleton	**15,290 ± 150**	Guerret, 1952; Gambier et al., 2000
Cap Blanc 1	France	F	Complete skeleton		von Bonin, 1935
Chancelade 1	France	M	Complete skeleton		Vallois, 1941–1946
La Madeleine 4	France	Child	Complete skeleton	**10,190 ± 100**	Heim, 1991; Gambier et al., 2000
Le Peyrat 5	France	M	Complete skeleton		Patte, 1968
Le Peyrat 6	France	F?	Postcranial bones		Patte, 1968
Montgaudier 4	France	Child	Cranium		Gambier, 1987
Roc-de-Cave	France	Adolescent F?	Incomplete skeleton	**11,210 ± 140**	Gambier et al., 2000; Bresson, 2000
Rond-du-Barry 8	France	M	Cranium	17,100 ± 450	Gambier and Houet, 1993
St. Germain-la-Rivière 1	France	F	Complete skeleton	**15,780 ± 200**	Blancard et al., 1972; Gambier et al., 2000
Dobritz 2	Germany	M?	Cranium		Grimm and Ullrich, 1965
Neuessing 2	Germany	M	Postcranial bones		Giesler, 1953
Oberkassel 1	Germany	M	Complete skeleton	**11,570 ± 110**	Henke, 1987; Orschiedt, 2000
Oberkassel 2	Germany	F	Complete skeleton	**12,180 ± 100**	Henke, 1987; Orschiedt, 2000
Arene Candide 2	Italy	M	Complete skeleton	10,000–11,000	Formicola et al., 2005; Paoli et al., 1980
Arene Candide 3	Italy	M	Skull	10,000–11,000	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 4	Italy	M	Skull	10,000–11,000	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 5	Italy	M	Complete skeleton	9,925 ± 50	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 6	Italy	Child	Complete skeleton	**9,925 ± 50**	Paoli et al., 1980; Formicola et al., 2005

(*continued*)

TABLE A1. (Continued)

Samples[a]	Country	Age/Sex	Remains	Date[b]	References
Arene Candide 8	Italy	Child	Complete skeleton	**10,655 ± 55**	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 10	Italy	M	Incomplete skeleton	10,000–11,000	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 11	Italy	Child	Complete skeleton	10,000–11,000	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 12	Italy	M	Skull	10,000–11,000	Paoli et al., 1980; Formicola et al., 2005
Arene Candide 16	Italy	Adolescent M	Incomplete skeleton	**10,810 ± 65**	Hedges, personal communication Paoli et al., 1980
Continenza 7	Italy	M	Incomplete skeleton	10,280 ± 110; 10,230 ± 110	Grifoni Cremonesi et al., 1995
Grotte des Enfants 1	Italy	Child	Complete skeleton	**11,130 ± 100**	Henry-Gambier, 2001
Grotte des Enfants 2	Italy	Child	Complete skeleton	11,130 ± 100	Henry-Gambier, 2001
Grotte des Enfants 3	Italy	F	Incomplete skeleton	<12,200 ± 400; 11,130 ± 100	Henry-Gambier, 2001; Thommeret and Thommeret 1973
Le Mura	Italy	Child	Complete skeleton		Alciati et al., 2005
Maritza 1	Italy	Child	Incomplete skeleton		Favati Vanni, 1964; Alciati et al., 2005
Maritza 2	Italy	M	Incomplete skeleton		Borgognini Tarli, 1969; Alciati et al., 2005
Ortucchio 1	Italy	F	Skull	12,619 ± 410	Parenti, 1960; Alciati et al., 2005
Romanelli 1	Italy	M	Complete skeleton		Alciati et al., 2005
Romanelli 2	Italy	Child	Incomplete skeleton		Alciati et al., 2005
Romanelli 3	Italy	Child	Incomplete skeleton		Alciati et al., 2005
Romito 1	Italy	F	Complete skeleton	11,150 ± 150	Mallegni and Fabbri, 1995
Romito 2	Italy	Adolescent M?	Complete skeleton	11,150 ± 150	Frayer et al., 1988; Mallegni and Fabbri, 1995
Romito 3	Italy	M	Complete skeleton	11,150 ± 150	Mallegni and Fabbri, 1995
Romito 4	Italy	F	Complete skeleton	11,150 ± 150	Mallegni and Fabbri, 1995
Romito 5	Italy	F	Complete skeleton	11,150 ± 150	Mallegni and Fabbri, 1995
Romito 6	Italy	M	Complete skeleton	11,150 ± 150	Mallegni and Fabbri, 1995
Romito 7	Italy	M	Complete skeleton	12,108 ± 50	Martini et al., 2004
Romito 8	Italy	M?	Complete skeleton	12,108 ± 50	Martini, 2006
San Teodoro 1	Italy	F	Complete skeleton		Graziosi, 1947; Fabbri, 1993
San Teodoro 2	Italy	?	Skull		Graziosi, 1947
San Teodoro 3	Italy	?	Incomplete skeleton		Graziosi, 1947
San Teodoro 4	Italy	F	Postcranial skeleton		Graziosi, 1947; Fabbri, 1993
Tagliente 2	Italy	M	Postcranial skeleton	13,430 ± 180; 12,040 ± 170	Corrain 1977; Alciati et al., 2005
Vado all' Arancio 1	Italy	M	Complete skeleton	11,330 ± 50; 11,600 ± 130	Minellono et al., 1980
Vado all' Arancio 2	Italy	Child	Incomplete skeleton	11,330 ± 50; 11,600 ± 130	Minellono et al., 1980
Villabruna 1	Italy	M	Complete skeleton	**12,140 ± 150**	Vercellotti et al., 2008
Kostenky 1	Russia	M	Complete skeleton		Debetz, 1955
Moča 1	Slovak Republic	F?	Skull	**11,255 ± 80**	Šefčáková et al., 1999
Roc del Migdia 1	Spain	F	Incomplete skeleton	11,520 ± 220	Turbón 1989; Garralda, 1991
Le Bichon 1	Switzerland	M	Complete skeleton	**11,610 ± 110; 11,760 ± 110**	Morel, 1993
Anetovka 1	Ukraine	?	Skull	18,040 ± 150; 19,710 ± 120	Ullrich, 1992
Africa					
Afalou	Algeria		Numerous individuals	13,120 ± 370; 11,450 ± 230	Boule et al., 1934; Chamla, 1978; Hachi, 1996

(*continued*)

TABLE A1. (Continued)

Samples[a]	Country	Age/Sex	Remains	Date[b]	References
Taforalt	Morroco		Numerous individuals	16,750	Ferembach, 1962
Tushka	Egypt		Numerous individuals	12,000–10,000	Wendorf, 1968
Wadi Halfa	Sudan		Numerous individuals (fragmentary)	11,950–6,400	Greene and Armelagos, 1972; Wendorf et al., 1965
Jebel Sahaba	Sudan		Numerous individuals (fragmentary)	13,740–600	Anderson, 1968; Wendorf et al., 1968
Asia (Western)					
Mugharet E1-Wad	Israel		Numerous individuals	Natufian	McCown 1939
Mugharet E1 - Kebara	Israel		Numerous individuals	Natufian	McCown 1939
Ein Gev 1	Israel	F	Incomplete skeleton	15,700 ± 415	Arensburg and Bar-Yosef, 1973
Neve David	Israel	?	Incomplete skeleton (fragmentary)	12,610 ± 130; 13,400 ± 180	Kaufman, 1988
Ohalo II	Israel	M	Complete skeleton	19,000	Nadel and Hershkovitz, 1991; Hershkovitz et al., 1995
Wadi Mataha (F-81)	Jordan	M	Complete skeleton	17,579–16,457	Stock et al., 2005
Asia (Eastern)					
Batadomba lena 1	Sri Lanka	M	Incomplete skeleton	15,830 ± 1,260	Kennedy et al., 1987; Kennedy and Deraniyagala, 1989
Batadomba lena 2	Sri Lanka	F	Incomplete skeleton (fragmentary)	15,830 ± 1,260	Kennedy et al., 1987; Kennedy and Deraniyagala, 1989
Beli Lena Kitulgala 1	Sri Lanka	F?	Incomplete skeleton (fragmentary)	12,260 ± 870	Kennedy et al., 1987
Beli Lena Kitulgala 2	Sri Lanka	Child	Incomplete skeleton (fragmentary)	12,260 ± 870	Kennedy et al., 1987
Tam Hang 20534	Laos	F	Complete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20535	Laos	F	Complete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20536	Laos	M	Incomplete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20537	Laos	F	Complete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20538	Laos	M	Complete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20540	Laos	F	Incomplete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20550	Laos	F	Incomplete skeleton	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20533	Laos	F	Cranium	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20542	Laos	M	Cranuim	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005
Tam Hang 20539	Laos	F	Cranuim	15,800	Fromaget and Saurin 1936; Demeter, 2000; Shackelford, 2005

[a] The appendix comprises individuals preserving at least the cranium or a largely complete postcranial sleleton, and reports sex and age attributions and absolute dates. Age is not specified for adults. Question mark in Age/Sex indicates that sex for specimen was not available or was uncertain.
[b] Dates are uncalibrated. Direct dates are in bold.

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